

A 3D brittle fracture model with effect of microstructure, strain gradient and strain rate

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ABSTRACT

Based on the two-scale asymptotic expansion theory, we establish a dynamic fracture model for 3D micro-cracked materials which extends the previous 2D results presented by the authors in Rao (2022); Rao et al. (2023). Using the two-scale theory for 3D problems, an analytical formulation of dynamic energy release rate is obtained that includes additive contributions of macroscopic strain, strain gradient and strain rate. The coefficients of the strain, strain gradient and strain rate are related to the microstructural size and derivatives of the homogenized (effective) elastodynamics moduli, which are determined by solutions of elementary elastodynamics problems defined in a reference unit cell. The microdamage evolution equation is developed by combining the analytical formulation of dynamic energy release rate with the Griffith fracture law. In contrast to the two-dimensional case where the normalized microcrack length is used as the measure of microdamage, in the three-dimensional case, the normalized microcrack area is used as the measure of microdamage, and then, the dynamic evolution equation of the microdamage variable for 3D problems has the same form as that for 2D cases. We analyze the properties of the homogenized elastodynamics moduli and compare them with those in the 2D cases. The coupling of microstructure size, strain gradient and strain rate are analyzed by examining local material responses and spallation experiment. The finite element simulations based on the model are well validated against available experimental results and previous reports.

1. Introduction

Brittle fracture is a size-dependent mechanical behavior in solids that has been verified by many experiments. It is influenced by the microstructure, such as voids, defects, dislocations, grains and grain boundaries, at different scales. The instability of an existing macroscopic crack is usually caused by the nucleation, growth and coalescence of microscopic cavities near the crack tip. Therefore, it is essential to consider the effect of microstructure to describe macroscopic fracture. Many attempts have been made to reveal the effect of microstructure on fracture to explain experimental observations [1–3]. Dynamic fracture, which is usually accompanied by the nucleation, propagation and interaction of micro-cracks as well as the initiation, propagation and branching of macro-cracks [4], is a rate-dependent and multi-scale phenomenon. It is a challenging research object due to the complex distribution of microstructure

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Nomenclature

E	Young's modulus
ν	Poisson's ratio
ρ	Material density
λ, μ	Lamé constants
Ω	Bounded region in 3D
L	Side length of a square crack surface in the reference cell
S	Area of square crack surface in the reference cell ($S = L^2$)
l	Side length of a square crack surface in the unit cell
s	Area of square crack surface in the unit cell ($s = l^2$)
Σ	Macroscopic stress tensor (Σ_{ij})
$\mathcal{E}$	Macroscopic strain tensor ($\mathcal{E}_{ij}$)
X	Spatial coordinates of macroscopic structure (X_i)
U	Macroscopic displacement field (U_i)
C, D, F	Homogenized elastic tensor (C_{ijkl} , D_{pqjmn} , F_{ijkl})
Σ_c	Critical stress
E	Strain energy density
W	Strain energy of the RVE
$\overline{W}$	Strain energy
T	Kinetic energy density
K	Kinetic energy of the RVE
$\overline{K}$	Kinetic energy
G	Energy release rate
G_c	Critical energy release rate (fracture toughness)
C_R	Rayleigh wave speed
ε	Microstructural scaling parameter
u^ε	Displacement field in homogenization model (u_i^ε)
σ^ε	Stress field in homogenization model (σ_{ij}^ε)
x, y	Spatial coordinates in homogenization model ($y = x/\varepsilon$)
δ	Kronecker symbol (δ_{ij})
RVE	Representative Volume Element
PMMA	Polymethyl methacrylate

as well as the complex loading ways. Understanding the scale effects and strain rate effects on brittle fracture has become one of the hot research directions nowadays. It still requires greater efforts to develop the corresponding cross-scale analysis methods.

The prediction of dynamic fracture for brittle and quasi-brittle materials, especially the propagation and interaction of micro-cracks and their impact on macro fracture, has drawn much attention in the domain of solid mechanics. Although many studies have been conducted to reveal the microstructure effects using micro-mechanical approaches [5–7], it remains a challenging issue to date. To model brittle fracture under compressive stress states, Ashby and Hallam [8] proposed a framework of damage mechanics based on critical stress criteria by considering the initial crack length, orientation and the coefficient of friction. Gambarotta and Lagomarsino [9] proposed a damage model to describe the different behaviors of brittle and quasi-brittle materials under tensile or compressive stresses by considering the unilateral friction contact between the micro-crack faces, which revealed anisotropic damage and loading path dependence for brittle fracture. Rafiee et al. [10] employed numerical investigation of micro-crack nucleation in front of a fast-running macro-crack and the numerical result indicated that fracture toughness increases with increasing crack speed. Based on two-scale asymptotic analysis and energy flux theory (J-integral) at the tip of a micro-crack, Dascalu et al. [11] and Keita et al. [12] developed a brittle fracture model to account for the size effect on macro behavior, such as the localization of damage, fracture strength, rate-dependent behavior and wave dispersion for dynamic loading. Markenscoff and Dascalu [13] analyzed the influence of micro-crack interaction on overall damage and showed that the singular stresses, as the cracks approach each other, can lead to damage amplification as a result of the singular interaction of micro-cracks.

The size effect becomes significant when the largest microstructural scale in a material approaches the relevant macroscopic structure scale. It is essential to develop a non-local mechanical model to describe the size effect for macroscopic mechanical response [14–16]. By introducing higher-order gradients of strain to the elastic constitutive equations, strain gradient elasticity has become a hot topic of research in solid mechanics theory [17–19]. The homogenization technique is an effective way to develop a strain gradient constitutive model for linear elastic heterogeneous materials with periodic microstructures. It has been used in many modeling studies and corresponding numerical results have indicated that the influence of the strain gradient rapidly decreases when the considered element becomes larger with respect to the size of the basic cell called RVE [20–22], which means that the influence of the strain gradient is negligible for homogeneous materials. For brittle solids with many micro-cracks inside, strain gradient damage models have been developed to assess the size effect of fracture and analyze the strain gradient effect on fracture

parameters such as fracture strength and energy release rate [23–26]. Apart from pure elasticity theory, the strain gradient theory is also applied to many other theories. In plasticity theory, it is used to analyze plastic deformation (e.g., strain hardening, flow theory, dislocation theory) [27–29]. In elastoplastic fracture theory, it is used to analyze the mechanics behavior around the crack tip [30,31]. Furthermore, strain gradient elasticity has been employed to regularize the singular stress field at the crack tip, yielding a more accurate and physically realistic stress distribution compared to the classical linear elastic solution. This approach has been used, for instance, to describe experimental observations using criteria based on the resulting regularized stresses [32,33].

The effect of strain rate is an essential factor for the extension of micro-cracks in dynamic fracture theory. Macro brittle fracture is sensitive to loading rate, especially under high-rate loading, as reported in many literature sources [12,34,35]. In fact, the micro inertia effect in the micro-crack tip is significant for higher loading rates. Reinhardt and Weerheijm [36] proposed a damage model to account for flaws in concrete and the inertia effects around a crack at high loading rates, verifying the increase of apparent tensile strength with increasing loading rate. To account for the micro inertia effect, Hai et al. [37] developed a phase-field damage model by introducing micro kinetic energy to the classical phase-field fracture model. It is shown that the micro inertia results in an increase in the resistance to failure and a decrease in crack propagation velocity. Shock-induced spalling fracture is a typical dynamic fracture with a high strain rate, which occurs when rarefaction waves resulting from the reflection of compression shock waves at stress-free surfaces interact to form tensile stresses [38]. It has become a standard way to probe dynamic material properties under extremely high strain rates. The rate-dependence for spall strength has been observed by spalling experiments [39–41] and the so-called Denoual–Forquin–Hild (DFH) model is used to simulate the spall fracture phenomenon with the effect of strain rate.

Although numerous fracture models have been proposed to predict dynamic crack propagation in 2D and significant progress has been made in theory and numerical simulation by researchers in revealing fracture mechanisms. In contrast, the fracture model in 3D is less explored due to the complex configuration and propagation path of cracks. Landis et al. [42] analyzed the secondary toughening mechanisms of concrete fracture energy. Su et al. [43] developed a 3D fracture model with Monte Carlo methods to analyze the fracture behavior of random heterogeneous materials. Mirsayar [44] proposed the concept of “effective critical distance” (ECD) to reveal the effect of mode mixity on the size of the crack-tip processing zone in three-dimensional brittle/quasi-brittle fractures. In recent years, phase-field theory has become an effective method to simulate dynamic crack propagation in 3D due to its ability to simulate complex microstructure evolution. Based on variational methods and energy minimization, it can simulate curvilinear fracture paths, branching and even crack coalescence without predefining cracks [45,46].

In the present work, we propose a new brittle fracture model in 3D based on a two-scale asymptotic analysis method. We establish a dynamic fracture criterion and a damage evolution equation that take into account the effect of strain gradient, strain rate and microstructure. This is a generalization of the authors’ previous work on the 2D case [47]. Instead of considering a Griffith crack in two-dimensional space, we consider an embedded micro-crack in a cell as the microstructure. Based on energy analysis and the Griffith fracture criterion, we rigorously derive the dynamic energy release rate of the cell containing a micro-crack using two-scale asymptotic analysis. We then establish the local dynamic evolution equation of the micro-crack by considering the speed-dependent critical fracture energy $G_c(v)$. With this model, it is easy to conduct a coupled analysis of strain gradient effects, strain rate effects and microstructure size effects on brittle fracture. It can also reveal the competition mechanism between these effects, which is helpful in further understanding the essence of brittle fracture.

The paper is organized as follows. In Section 2, the two-scale framework involving micro-inertia effects is established. The so-called cell equation and the homogenized equilibrium equation are obtained by two-scale asymptotic analysis. In Section 3, the model for micro-crack propagation and macro fracture is rigorously derived through energy analysis based on the two-scale asymptotic analysis. In Section 4, numerical results for local response and macro fracture are proposed, the comparisons with experiments are also given to verify the effectiveness of the model. The work is concluded in Section 5. Throughout the paper, the Einstein summation convention for repeated indices is adopted and all subscripts and superscripts in this paper take values 1, 2, 3.

2. Modeling

2.1. Macro problem

We consider a bounded elastic solid Ω containing a macro-crack in three-dimensional space, which features a microstructure with numerous randomly distributed embedded micro-cracks. For simplicity, we assume that the micro-cracks exhibit a locally periodic distribution (see Fig. 1). The average distance between two adjacent micro-cracks is denoted by ε , which can also be interpreted as the period. A cell containing a squared micro-crack with side length ε is denoted by Y_ε . The side length of the micro-crack surface is represented by L and the area of the micro-crack surface is calculated as $S = L^2$.

The macroscopic momentum equation can be expressed as:

$$\frac{\partial \Sigma_{ij}}{\partial X_j} = \rho \frac{\partial^2 U_i}{\partial t^2} \quad \text{in } \Omega, \quad (1)$$

where ρ is the density of the elastic solid, U is the displacement field, $\{\Sigma_{ij}\}$ is macro stress tensor and X is coordinates in the macro structure. Before failure, the stress–strain relation is

$$\Sigma_{ij} = C_{ijkl} \mathcal{E}_{kl}, \quad (2)$$

where $\{C_{ijkl}\}$ is the homogenized elastic tensor which depends on microstructure and $\mathcal{E}$ is macro strain tensor that can be expressed as:

$$\mathcal{E}_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial X_j} + \frac{\partial U_j}{\partial X_i} \right). \quad (3)$$

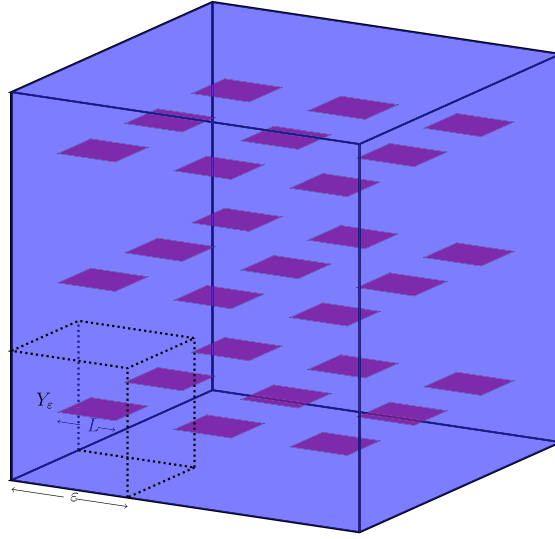


Fig. 1. Local periodic micro-cracks assumption.

2.2. Two-scale asymptotic analysis

The infinite periodic Representative Volume Element (RVE) is denoted as $B(= \mathbb{R}^3)$, while the region occupied by micro-cracks is denoted as C . In the solid portion of the domain, denoted as $B_s = B \setminus C$, the equilibrium equation can be expressed as:

$$\frac{\partial \sigma_{ij}^\epsilon}{\partial x_j} = \rho \frac{\partial^2 u_i^\epsilon}{\partial t^2} \quad \text{in } B_s, \quad (4)$$

and the linear elastic constitutive relation is

$$\sigma_{ij}^\epsilon = a_{ijkl} e_{kl}(\mathbf{u}^\epsilon), \quad (5)$$

where $\mathbf{u}^\epsilon$, σ^ϵ and $e_{ij}(\mathbf{u}^\epsilon)$ represent displacement, stress and strain fields respectively and

$$e_{ij}(\mathbf{u}^\epsilon) = \frac{1}{2} \left(\frac{\partial u_i^\epsilon}{\partial x_j} + \frac{\partial u_j^\epsilon}{\partial x_i} \right). \quad (6)$$

Assuming isotropic elasticity of the solid material within the RVE, we can express the equilibrium equation as,

$$a_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (7)$$

where λ and μ are Lamé constants, δ_{ij} is the *Kronecker symbol*.

We consider the possibility of crack extension and neglect crack closure, thereby assuming a fracture mode of Mode-I. As a result, we adopt the following unstressed boundary condition at the surfaces of the crack:

$$\sigma^\epsilon \cdot \mathbf{n} = \mathbf{0}, \quad (8)$$

where $\mathbf{n}$ is the outward unit normal vector on crack surfaces.

To facilitate analysis, we use a scale transformation $\mathbf{y} = \mathbf{x}/\epsilon$, which maps the periodic cell Y_ϵ to a unit reference cell $Y = [0, 1] \times [0, 1] \times [0, 1]$ (as shown in Fig. 2). The side length of a squared micro-crack surface in the reference cell is $l = L/\epsilon$ and the area is calculated as $s = l^2 = S/\epsilon^2$. These measures represent the degree of damage in a local cell, where $s = 0$ (or $l = 0$) represents a state without any damage, while $s = 1$ (or $l = 1$) means that the material is fully fractured locally.

Correspondingly, the crack C is mapped to C_Y , which has a length of l in Y . The upper and lower surfaces of the crack are denoted as C_Y^+ and C_Y^- , respectively. The solid portion in the reference cell is denoted as $Y_s = Y \setminus C_Y$.

For a given scalar function $f^\epsilon(\mathbf{x}) = f(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}, \mathbf{x}/\epsilon)$, we can apply the chain rule of derivatives as follows:

$$\left. \frac{\partial f^\epsilon}{\partial x_j} \right|_{(\mathbf{x})} = \left. \frac{\partial f}{\partial x_j} \right|_{(\mathbf{x}, \frac{\mathbf{x}}{\epsilon})} + \frac{1}{\epsilon} \left. \frac{\partial f}{\partial y_j} \right|_{(\mathbf{x}, \frac{\mathbf{x}}{\epsilon})}. \quad (9)$$

According to the asymptotic homogenization theory [48], the RVE solution $\mathbf{u}^\epsilon$ has the following series form:

$$\mathbf{u}^\epsilon(\mathbf{x}, t) = \mathbf{u}^{(0)}(\mathbf{x}, \mathbf{y}, t) + \epsilon \mathbf{u}^{(1)}(\mathbf{x}, \mathbf{y}, t) + \epsilon^2 \mathbf{u}^{(2)}(\mathbf{x}, \mathbf{y}, t) + \dots, \quad (10)$$

where $\mathbf{u}^{(i)}(\mathbf{x}, \mathbf{y}, t)$, ($i \in \mathbb{N}$) are Y -periodic about micro variable $\mathbf{y}$.

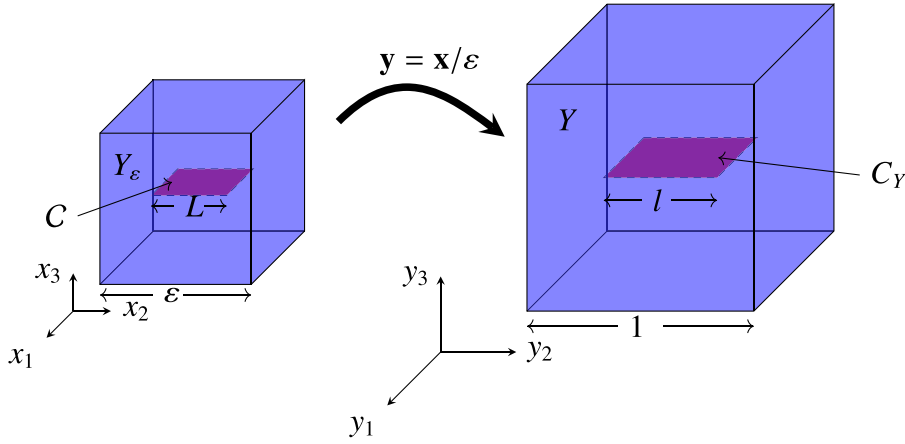


Fig. 2. Scale transformation.

For simplicity, we introduce the following notations for the symmetric parts of spatial derivatives of a vector function $\mathbf{f}(\mathbf{x}, \mathbf{y})$:

$$e_{xij}(\mathbf{f}) = \frac{1}{2} \left(\frac{\partial f_i}{\partial x_j} + \frac{\partial f_j}{\partial x_i} \right), \quad e_{yij}(\mathbf{f}) = \frac{1}{2} \left(\frac{\partial f_i}{\partial y_j} + \frac{\partial f_j}{\partial y_i} \right),$$

and define the following vector function space:

$$D(Y_s) = \left\{ \mathbf{u} \mid \mathbf{u} \in [H^1(Y_s)]^3, \mathbf{u} \text{ is } Y\text{-periodic in } \mathbf{y}, \int_{Y_s} \mathbf{u} \, d\mathbf{y} = \mathbf{0} \right\}.$$

where $H^1(Y_s)$ denotes the Sobolev space [48].

By substituting the asymptotic expansion (10) into (4) and gathering the terms for each order of ε , we obtain the following partial differential equations defined in the unit cell:

$$\begin{cases} \frac{\partial}{\partial y_j} (a_{ijkl} e_{ykl}(\mathbf{u}^{(0)})) = 0, & \text{in } Y_s, \\ a_{ijkl} e_{ykl}(\mathbf{u}^{(0)}) n_j = 0, & \text{on } C_Y^\pm; \end{cases} \quad (11)$$

$$\begin{cases} \frac{\partial}{\partial y_j} (a_{ijkl} (e_{xkl}(\mathbf{u}^{(0)}) + e_{ykl}(\mathbf{u}^{(1)}))) = 0, & \text{in } Y_s, \\ a_{ijkl} (e_{ykl}(\mathbf{u}^{(1)}) + e_{xkl}(\mathbf{u}^{(0)})) n_j = 0, & \text{on } C_Y^\pm; \end{cases} \quad (12)$$

$$\begin{cases} \frac{\partial}{\partial x_j} a_{ijkl} (e_{xkl}(\mathbf{u}^{(0)}) + e_{ykl}(\mathbf{u}^{(1)})) + \frac{\partial}{\partial y_j} a_{ijkl} (e_{xkl}(\mathbf{u}^{(1)}) + e_{ykl}(\mathbf{u}^{(2)})) = \rho \frac{\partial^2 u_i^{(0)}}{\partial t^2}, & \text{in } Y_s, \\ a_{ijkl} (e_{xkl}(\mathbf{u}^{(1)}) + e_{ykl}(\mathbf{u}^{(2)})) n_j = 0, & \text{on } C_Y^\pm. \end{cases} \quad (13)$$

According to (11), $\mathbf{u}^{(0)}$ is independent of the microscopic variable $\mathbf{y}$, i.e., $\mathbf{u}^{(0)} = \mathbf{u}^{(0)}(\mathbf{x}, t)$ [48].

As for $\mathbf{u}^{(1)}$, we can find a variable separative solution with the following form:

$$\mathbf{u}^{(1)}(\mathbf{x}, \mathbf{y}, t) = \mathbf{N}^{pq}(\mathbf{y}) e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t)), \quad (14)$$

where $\mathbf{N}^{pq}(\mathbf{y}) \in D(Y_s)$, $(p, q = 1, 2, 3)$ are the first-order cell solutions satisfying:

$$\begin{cases} \frac{\partial}{\partial y_j} (a_{ijkl} e_{ykl}(\mathbf{N}^{pq}) + a_{ijpq}) = 0, & \text{in } Y_s, \\ (a_{ijkl} e_{ykl}(\mathbf{N}^{pq}) + a_{ijpq}) n_j = 0, & \text{on } C_Y^\pm. \end{cases} \quad (15)$$

We can directly obtain $\mathbf{N}^{pq}(\mathbf{y}) = \mathbf{N}^{qp}(\mathbf{y})$ from (15) with the symmetry of tensor $\{a_{ijkl}\}$.

Integrating (13) over Y_s , we obtain:

$$\frac{\partial}{\partial x_j} \int_{Y_s} a_{ijkl} (e_{xkl}(\mathbf{u}^{(0)}) + e_{ykl}(\mathbf{u}^{(1)})) \, d\mathbf{y} = \int_{Y_s} \rho \frac{\partial^2 u_i^{(0)}}{\partial t^2} \, d\mathbf{y}. \quad (16)$$

By substituting (14) into (16), we obtain:

$$\frac{\partial}{\partial x_j} (C_{ijkl} e_{xkl}(\mathbf{u}^{(0)})) = \rho \frac{\partial^2 u_i^{(0)}}{\partial t^2}, \quad (17)$$

where

$$C_{ijkl} = \int_{Y_s} a_{ijkl} + a_{ijmn} e_{ymn}(\mathbf{N}^{kl}) \, dy \quad (18)$$

is the homogenized elastic tensor. Due to the symmetry of cell function and geometric configuration, $\mathbf{C}$ has only six non-zero independent components (see [Appendix B](#)).

Eq. (17) is referred to as the homogenization equation. It has been demonstrated that each component C_{ijkl} is dependent on Y_s by means of the integral region and cell function $\mathbf{N}^{kl}$, as defined in Eq. (18). Consequently, C_{ijkl} is contingent upon the damage variable l , but is independent of the microstructure parameter ε .

Remark 1. In the 3D framework of this paper, the axis labels differ from those used in the previous 2D work [47]. In particular, the direction perpendicular to the crack plane is denoted as x_3 (or y_3) here, whereas the corresponding direction was labeled x_2 in [47].

3. Macroscopic formulation

3.1. Brittle fracture criterion based on asymptotic analysis

In the following sections, we make the assumption that the crack extends in a linear fashion along its original direction at the same rate in all four directions.

Firstly, we provide an energy analysis for a cell that contains a single micro-crack. The local strain energy density E within the periodic representative volume element(RVE) can be expressed as:

$$E = \frac{1}{2} \boldsymbol{\sigma}^\varepsilon : \mathbf{e}(\mathbf{u}^\varepsilon) = \frac{1}{2} a_{ijkl} e_{ij}(\mathbf{u}^\varepsilon) e_{kl}(\mathbf{u}^\varepsilon). \quad (19)$$

Taking the following first-order approximation $O(\varepsilon)$ in (10):

$$\mathbf{u}^\varepsilon(\mathbf{x}, t) = \mathbf{u}^{(0)}(\mathbf{x}, t) + \varepsilon \mathbf{u}^{(1)}(\mathbf{x}, \mathbf{y}, t) = \mathbf{u}^{(0)}(\mathbf{x}, t) + \varepsilon \mathbf{N}^{pq}(\mathbf{y}) e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t)). \quad (20)$$

We obtain:

$$e_{ij}(\mathbf{u}^\varepsilon)(\mathbf{x}) = e_{xij}(\mathbf{u}^{(0)})(\mathbf{x}) + e_{yij}(\mathbf{N}^{pq})(\mathbf{y}) e_{xpq}(\mathbf{u}^{(0)})(\mathbf{x}) + \frac{\varepsilon}{2} \left(\mathbf{N}_i^{pq}(\mathbf{y}) \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j}(\mathbf{x}) + \mathbf{N}_j^{pq}(\mathbf{y}) \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_i}(\mathbf{x}) \right). \quad (21)$$

By noting the symmetry of a_{ijkl} and substituting (10) into (19), we obtain:

$$\begin{aligned} E = & \frac{1}{2} a_{ijkl} e_{xij}(\mathbf{u}^{(0)}) e_{xkl}(\mathbf{u}^{(0)}) + \frac{1}{2} a_{ijkl} \mathbf{N}_{i,j}^{pq} \mathbf{N}_{k,l}^{mn} e_{xpq}(\mathbf{u}^{(0)}) e_{xmn}(\mathbf{u}^{(0)}) + a_{ijkl} \mathbf{N}_{i,j}^{pq} e_{xpq}(\mathbf{u}^{(0)}) e_{xkl}(\mathbf{u}^{(0)}) \\ & + \varepsilon a_{ijkl} \left(e_{xij}(\mathbf{u}^{(0)}) + \mathbf{N}_{i,j}^{pq} e_{xpq}(\mathbf{u}^{(0)}) \right) \mathbf{N}_k^{mn} \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} + \frac{\varepsilon^2}{2} a_{ijkl} \mathbf{N}_i^{pq} \mathbf{N}_k^{mn} \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l}. \end{aligned} \quad (22)$$

By integrating (22) in Y_ε , we obtain the total strain energy of the unit cell:

$$W = \int_{Y_\varepsilon} E \, d\mathbf{x}.$$

Noting that $\mathbf{u}^{(0)}$ and its derivatives are slowly varying functions over the local domain Y_ε , for any continuous functions $F_1(\mathbf{x})$, $F_2(\mathbf{x})$, and $G(\mathbf{x})$, the following integral approximation holds:

$$\int_{Y_\varepsilon} G(\mathbf{x}) F_1(\mathbf{u}^{(0)}) F_2(\mathbf{u}_{i,j}^{(0)}) \, d\mathbf{x} \approx F_1(\langle \mathbf{u}^{(0)} \rangle) F_2(\langle \mathbf{u}_{i,j}^{(0)} \rangle) \int_{Y_\varepsilon} G(\mathbf{x}) \, d\mathbf{x}, \quad (23)$$

where $\langle \cdot \rangle = \int_{Y_\varepsilon} \cdot \, d\mathbf{x} / |Y_\varepsilon|$ denotes the averaging operator over the domain Y_ε . So we can approximate the strain energy W as:

$$\begin{aligned} W \approx & \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xmn}(\mathbf{u}^{(0)}) \rangle \int_{Y_\varepsilon} \frac{1}{2} a_{ijkl} \mathbf{N}_{i,j}^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) \mathbf{N}_{k,l}^{mn} \left(\frac{\mathbf{x}}{\varepsilon} \right) \, d\mathbf{x} + \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle \int_{Y_\varepsilon} a_{ijkl} \mathbf{N}_{i,j}^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) \, d\mathbf{x} \\ & + \frac{\varepsilon^3}{2} a_{ijkl} \langle e_{xij}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle + \frac{\varepsilon^2}{2} \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} \right\rangle \int_{Y_\varepsilon} a_{ijkl} \mathbf{N}_i^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) \mathbf{N}_k^{mn} \left(\frac{\mathbf{x}}{\varepsilon} \right) \, d\mathbf{x}, \end{aligned} \quad (24)$$

In (24), the term corresponding to ε is zero due to the symmetry and anti-symmetry properties of the first-order cell solutions (see detailed proof in Appendix A). Using the variable substitution $\mathbf{y} = \mathbf{x}/\varepsilon$, we obtain:

$$\begin{aligned}
 W &= \frac{\varepsilon^3}{2} a_{ijkl} \langle e_{xij}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle + \frac{\varepsilon^3}{2} \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xmn}(\mathbf{u}^{(0)}) \rangle \int_{Y_s} a_{ijkl} \mathbf{N}_i^{pq}(\mathbf{y}) \mathbf{N}_{k,l}^{mn}(\mathbf{y}) \, d\mathbf{y} \\
 &+ \varepsilon^3 \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle \int_{Y_s} a_{ijkl} \mathbf{N}_i^{pq}(\mathbf{y}) \, d\mathbf{y} + \frac{\varepsilon^5}{2} \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} \right\rangle \int_{Y_s} a_{ijkl} \mathbf{N}_i^{pq}(\mathbf{y}) \mathbf{N}_k^{mn}(\mathbf{y}) \, d\mathbf{y} \\
 &= \frac{\varepsilon^3}{2} \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle \int_{Y_s} (a_{pqkl} + a_{kl ij} \mathbf{N}_i^{pq}(\mathbf{y})) \, d\mathbf{y} \\
 &+ \frac{\varepsilon^3}{2} \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xmn}(\mathbf{u}^{(0)}) \rangle \int_{Y_s} (a_{klpq} + a_{kl ij} \mathbf{N}_i^{pq}(\mathbf{y})) \mathbf{N}_{k,l}^{mn}(\mathbf{y}) \, d\mathbf{y} \\
 &+ \frac{\varepsilon^5}{2} \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} \right\rangle \int_{Y_s} a_{ijkl} \mathbf{N}_i^{pq}(\mathbf{y}) \mathbf{N}_k^{mn}(\mathbf{y}) \, d\mathbf{y} \\
 &= \frac{\varepsilon^3}{2} C_{pqkl} \langle e_{xpq}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle + \frac{\varepsilon^5}{2} D_{pqjmn} \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} \right\rangle,
 \end{aligned} \tag{25}$$

where

$$D_{pqjmn} = \int_{Y_s} a_{ijkl} \mathbf{N}_i^{pq}(\mathbf{y}) \mathbf{N}_k^{mn}(\mathbf{y}) \, d\mathbf{y}. \tag{26}$$

Similar to tensor $\mathbf{C}$, each component of $\mathbf{D}$ depends on the damage variable d . The symmetry of a_{ijkl} and the first-order cell solutions $\mathbf{N}^{pq}$ lead to the symmetry $D_{pqjmn} = D_{qpjmn} = D_{mnlpqj}$. It can be proved that there are only 21 independent components of $\mathbf{D}$ (see Appendix B).

In deriving (25), the equality

$$\int_{Y_s} (a_{klpq} + a_{kl ij} \mathbf{N}_i^{pq}(\mathbf{y})) \mathbf{N}_{k,l}^{mn}(\mathbf{y}) \, d\mathbf{y} = \int_{Y_s} (a_{klpq} + a_{kl ij} e_{yij}(\mathbf{N}^{pq})) e_{ykl}(\mathbf{N}^{mn}) \, d\mathbf{y} = 0,$$

is adopted, which is obtained from the weak form of first-order function $\mathbf{N}^{pq}(\mathbf{y})$.

Now we consider the kinetic energy in one cell. The kinetic energy density can be written as follows:

$$T = \frac{1}{2} \rho \frac{\partial \mathbf{u}^\varepsilon}{\partial t} \frac{\partial \mathbf{u}^\varepsilon}{\partial t}.$$

Then, the kinetic energy in one cell is given by integrating the above expression over the cell volume Y_ε :

$$K = \int_{Y_\varepsilon} T \, d\mathbf{x} = \int_{Y_\varepsilon} \frac{1}{2} \rho \frac{\partial \mathbf{u}^\varepsilon}{\partial t} \frac{\partial \mathbf{u}^\varepsilon}{\partial t} \, d\mathbf{x}. \tag{27}$$

Substituting (20) into (27), then K can be rewritten as follows:

$$\begin{aligned}
 K &= \int_{Y_\varepsilon} \frac{1}{2} \rho \frac{\partial \left(\mathbf{u}^{(0)}(\mathbf{x}, t) + \varepsilon \mathbf{N}^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t)) \right)}{\partial t} \cdot \frac{\partial \left(\mathbf{u}^{(0)}(\mathbf{x}, t) + \varepsilon \mathbf{N}^{mn} \left(\frac{\mathbf{x}}{\varepsilon} \right) e_{xmn}(\mathbf{u}^{(0)}(\mathbf{x}, t)) \right)}{\partial t} \, d\mathbf{x} \\
 &= \int_{Y_\varepsilon} \frac{1}{2} \rho \frac{\partial \mathbf{u}^{(0)}}{\partial t} \frac{\partial \mathbf{u}^{(0)}}{\partial t} \, d\mathbf{x} + \int_{Y_\varepsilon} \varepsilon \rho \frac{\partial \mathbf{u}^{(0)}}{\partial t} \cdot \frac{\partial \left(\mathbf{N}^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t)) \right)}{\partial t} \, d\mathbf{x} \\
 &+ \int_{Y_\varepsilon} \frac{\varepsilon^2}{2} \rho \frac{\partial \left(\mathbf{N}^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t)) \right)}{\partial t} \cdot \frac{\partial \left(\mathbf{N}^{mn} \left(\frac{\mathbf{x}}{\varepsilon} \right) e_{xmn}(\mathbf{u}^{(0)}(\mathbf{x}, t)) \right)}{\partial t} \, d\mathbf{x} \\
 &\approx \frac{\varepsilon^3}{2} \rho \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle + \varepsilon \rho \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle \int_{Y_\varepsilon} \mathbf{N}_i^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) \, d\mathbf{x} \\
 &+ \frac{\varepsilon^2}{2} \rho \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle \int_{Y_\varepsilon} \mathbf{N}_i^{pq} \left(\frac{\mathbf{x}}{\varepsilon} \right) \mathbf{N}_i^{mn} \left(\frac{\mathbf{x}}{\varepsilon} \right) \, d\mathbf{x} \\
 &\stackrel{\mathbf{y}=\mathbf{x}/\varepsilon}{=} \frac{\varepsilon^3}{2} \rho \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle + \frac{\varepsilon^5}{2} \rho \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle \int_{Y_s} \mathbf{N}_i^{pq}(\mathbf{y}) \mathbf{N}_i^{mn}(\mathbf{y}) \, d\mathbf{y} \\
 &= \frac{\varepsilon^3}{2} \rho \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle \left\langle \frac{\partial u_i^{(0)}}{\partial t} \right\rangle + F_{pqmn} \frac{\varepsilon^5}{2} \rho \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)}(\mathbf{x}, t))}{\partial t} \right\rangle,
 \end{aligned} \tag{28}$$

where

$$F_{pqmn} = \int_{Y_s} \mathbf{N}_i^{pq}(\mathbf{y}) \mathbf{N}_i^{mn}(\mathbf{y}) \, d\mathbf{y}. \tag{29}$$

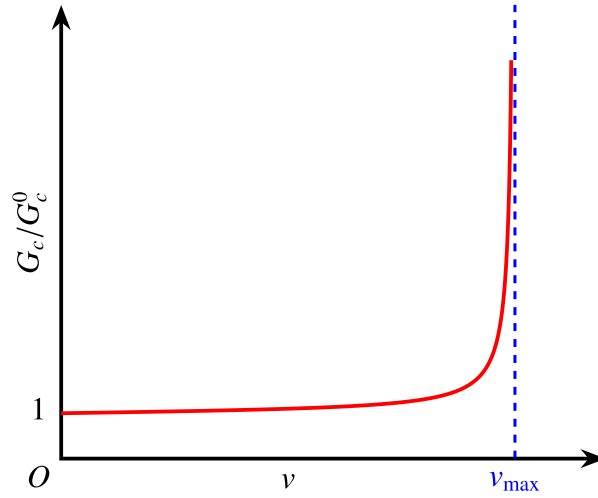


Fig. 3. Classical dependency relationship between G_c and v .

Due to the symmetry of the first-order cell functions $\mathbf{N}^{pq}(\mathbf{y})$, it is easy to see that each component of $\mathbf{F}$ depends on the damage variable d and $F_{pqmn} = F_{qpmn} = F_{mnpq}$ due to the symmetry of the first-order cell solutions from (29). These, together with the symmetry with respect to the plane $y_1 = y_2$, give five non-zero independent components (see Appendix B) of $\mathbf{F}$, listed as follows:

$$F_{1111}, F_{3333}, F_{1133}, F_{1212}, F_{1313}.$$

According to the Griffith criterion, for a cell containing a micro-crack, the micro-crack will extend if the energy release rate G exceeds a material-specific critical value G_c :

$$G > G_c, \quad (30)$$

so the micro-crack is stable if $G \leq G_c$.

In fact, the critical energy release rate G_c depends on the crack-tip speed v for dynamic fracture and is denoted by $G_c(v)$ with respect to v as shown in Fig. 3 [49,50]. It is shown that $G_c(v)$ has a weak dependence on growth rate of micro-crack v at the initial low-speed stage of micro-crack development and it increases significant as v is large enough. For the case of $v = 0$, the dynamic fracture degenerates into quasi-static fracture and $G_c(0) = G_c^0$. Note that the crack-tip speed cannot exceed Rayleigh wave speed C_R for Mode-I crack, which can be approximated as follows [51]:

$$C_R \approx \frac{0.862 + 1.14\nu}{1 + \nu} \sqrt{\frac{E}{2\rho(1 + \nu)}},$$

Here, ν is the Poisson ratio and E is the Young's modulus. We can use the following expression to approximate the functional relation between G_c and v , as shown in Fig. 3:

$$\frac{G_c(v)}{G_c^0} = 1 + \lambda \tan\left(\frac{\pi v}{2C_R}\right), \quad (31)$$

where λ is a material-specific parameter. In the following sections, $\lambda = 0.1$ is adopted. This choice can produce a limit speed $v_{\max} = C_R$ for Mode-I crack, as shown in Fig. 3, since the energy release rate vanishes for $v > C_R$ [49].

It should be noted that a more accurate function for $G_c(v)$ can be determined experimentally for specific materials. However, this topic is beyond the main purpose of the present work. In future work, a more accurate form of $G_c(v)$ for different materials may be used.

We can consider the variation of the mechanical energy with respect to the increment of the normalized micro-crack area as follows:

$$\begin{aligned} \delta(W + K) = & \frac{\varepsilon^3}{2} C'_{ijkl}(s) \delta s \langle e_{xij}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle + \frac{\varepsilon^5}{2} D'_{pqjmn}(s) \delta s \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} \right\rangle \\ & + \frac{\varepsilon^5}{2} \rho F'_{pqmn}(s) \delta s \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial t} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial t} \right\rangle. \end{aligned} \quad (32)$$

Based on the virtual crack extension technique, the dynamic energy release rate G can be calculated as follows [52]:

$$G = -\frac{d(W + K)}{dS}. \quad (33)$$

With the substitution $s = S/\epsilon^2$, we obtain:

$$\frac{d(\cdot)}{dS} = \frac{1}{\epsilon^2} \frac{d(\cdot)}{ds}. \quad (34)$$

The averaging of strain and strain gradients over a periodic cell can be considered as the macroscopic counterparts, i.e.,

$$\mathcal{E}_{ij} = \langle e_{xij}(\mathbf{u}^{(0)}) \rangle, \quad \mathcal{E}_{ij,k} = \left\langle \frac{\partial e_{xij}(\mathbf{u}^{(0)})}{\partial x_k} \right\rangle, \quad \dot{\mathcal{E}}_{ij} = \left\langle \frac{\partial e_{xij}(\mathbf{u}^{(0)})}{\partial t} \right\rangle. \quad (35)$$

The expression for the energy release rate can be reformulated as follows:

$$\begin{aligned} G &= -\frac{\epsilon}{2} \frac{dC_{ijkl}(s)}{ds} \langle e_{xij}(\mathbf{u}^{(0)}) \rangle \langle e_{xkl}(\mathbf{u}^{(0)}) \rangle - \frac{\epsilon^3}{2} \frac{dD_{pqjmn}(s)}{ds} \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial x_j} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial x_l} \right\rangle \\ &\quad - \frac{\epsilon^3}{2} \rho \frac{dF_{pqmn}(s)}{ds} \left\langle \frac{\partial e_{xpq}(\mathbf{u}^{(0)})}{\partial t} \right\rangle \left\langle \frac{\partial e_{xmn}(\mathbf{u}^{(0)})}{\partial t} \right\rangle \\ &= -\frac{\epsilon}{2} \frac{dC_{ijkl}(s)}{ds} \mathcal{E}_{ij} \mathcal{E}_{kl} - \frac{\epsilon^3}{2} \frac{dD_{pqjmn}(s)}{ds} \mathcal{E}_{pq,j} \mathcal{E}_{mn,l} - \frac{\epsilon^3}{2} \rho \frac{dF_{pqmn}(s)}{ds} \dot{\mathcal{E}}_{pq} \dot{\mathcal{E}}_{mn}. \end{aligned} \quad (36)$$

Note that $v = \frac{\dot{I}}{2} = \frac{\epsilon \dot{I}}{2}$ due to the former assumption, we obtain fracture criterion involving the effects of microstructure(ϵ and I), strain gradient effect and strain rate effect as follows:

$$-\frac{\epsilon}{2} C'_{ijkl}(s) \mathcal{E}_{ij} \mathcal{E}_{kl} - \frac{\epsilon^3}{2} D'_{pqjmn}(s) \mathcal{E}_{pq,j} \mathcal{E}_{mn,l} - \frac{\epsilon^3}{2} \rho F'_{pqmn}(s) \dot{\mathcal{E}}_{pq} \dot{\mathcal{E}}_{mn} = G_c(v) = G_c\left(\frac{\epsilon \dot{I}}{2}\right). \quad (37)$$

Here, the superscript ' represents derivatives with respect to s , i.e., $C'_{ijkl}(s) = \frac{dC_{ijkl}(s)}{ds}$, $D'_{pqjmn}(s) = \frac{dD_{pqjmn}(s)}{ds}$, $F'_{ijkl}(s) = \frac{dF_{ijkl}(s)}{ds}$.

We would like to emphasize that when $\dot{\mathcal{E}}_{pq} = 0$ and $v = 0$, the dynamic fracture criterion can be reduced to the quasi-static case. The coefficients of the obtained fracture criterion can be expressed as derivatives of effective elastic tensors, namely $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{F}$, with respect to the normalized damage variable s . The expressions for C_{ijkl} and D_{pqjmn} can be obtained using $\mathbf{N}^{pq}$ as per Eqs. (18) and (26). Although the derivation process is intricate, the final form of the fracture criterion in (37) is concise, involving terms related to strain, strain gradient and strain rate.

Notably, while the overall form of the fracture criterion shares a structural similarity with the 2D case in [47], the transition to 3D introduces a richer tensorial structure. The damage variable is now based on crack area, and the effective tensors $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{F}$ acquire additional independent components associated with the two in-plane principal directions of the crack surface. These features substantially modify the degradation laws and the coupling between strain, strain gradient, and strain rate, resulting in a fracture criterion with characteristics unique to the 3D setting.

3.2. Governing equations

The macroscopic behavior of the homogenized medium is characterized by its dependence on strain, strain gradients, and strain rates. The corresponding governing equations can be obtained directly from Hamilton's principle applied to the effective kinetic and potential energy densities.

The kinetic energy contains the classical inertia term and an additional micro-inertia contribution associated with the rate of strain $\bar{K}$ can be written as:

$$\bar{K} = \frac{1}{|Y_\epsilon|} \int_\Omega K d\mathbf{X} = \int_\Omega \left(\frac{1}{2} \rho \dot{U}_i \dot{U}_i + \frac{\epsilon^2}{2} F_{ijkl} \dot{\mathcal{E}}_{ij} \dot{\mathcal{E}}_{kl} \right) d\mathbf{X},$$

The strain energy is written as a quadratic function of the strain $\mathcal{E}_{ij}$ and the strain gradient $\mathcal{E}_{ij,k}$:

$$\bar{W} = \frac{1}{|Y_\epsilon|} \int_\Omega W d\mathbf{X} = \int_\Omega \left(\frac{1}{2} C_{ijkl} \mathcal{E}_{ij} \mathcal{E}_{kl} + \frac{\epsilon^2}{2} D_{pqjmn} \mathcal{E}_{pq,j} \mathcal{E}_{mn,l} \right) d\mathbf{X}.$$

External work is taken as:

$$W_{\text{ext}} = \int_\Omega f_i U_i d\mathbf{X} + \int_{\Gamma_t} g_i U_i d\Gamma + \int_{\Gamma_q} r_i U_{i,n} d\Gamma,$$

where $U_{i,n} = n_k U_{i,k}$ and n_k is the outward unit normal.

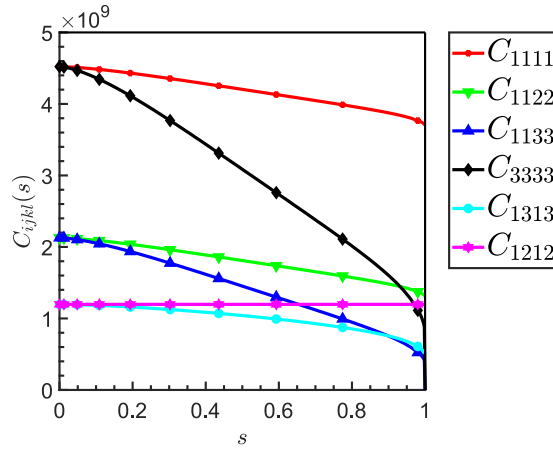
Hamilton's principle requires the stationarity of the action:

$$\delta \int_{t_0}^{t_1} (\bar{K} - \bar{W} + W_{\text{ext}}) dt = 0, \quad (38)$$

for all admissible variations δU_i that vanish at $t = t_0, t_1$ and on Γ_U .

We introduce the stress tensor Σ_{ij} and the double stress tensor Γ_{ijk} as the energy conjugates of the strain and its gradient as follows.

$$\sigma_{ij} = C_{ijkl} \mathcal{E}_{kl}, \quad \Gamma_{ijk} = \epsilon^2 D_{ijkmn} \mathcal{E}_{mn,l}.$$

Fig. 4. $C_{ijkl} - s$ curve.

For a detailed discussion, the reader is referred to Li et al. [53]; to ensure structural integrity, we present the final governing equations as follows:

$$\left\{ \begin{array}{l} \rho \ddot{U}_i - \varepsilon^2 F_{ijkl} \ddot{\varepsilon}_{kl,j} - \frac{\partial \Sigma_{ij}}{\partial X_j} + \varepsilon^2 \frac{\partial^2}{\partial X_k \partial X_j} (\Gamma_{ijk}) = f_i, \text{ in } \Omega, \\ U = \bar{U} \text{ or } n_j \left(\Sigma_{ij} - \frac{\partial \Gamma_{ijk}}{\partial X_i} \right) + n_i n_j \Gamma_{ijk} (D_l n_l) - D_j (n_i \Gamma_{ijk}) + \varepsilon^2 F_{ijkl} \left(\frac{\partial \ddot{U}_i}{\partial X_j} \right) n_l = g_k, \text{ on } \partial \Omega, \\ \frac{\partial U}{\partial n} = \frac{\partial \bar{U}}{\partial n} \text{ or } n_i n_j \Gamma_{ijk} = r_i, \text{ on } \partial \Omega, \\ U = \bar{U} \text{ or } [n_i k_j \Gamma_{ijk}] = p_k, \text{ on } C_m, \\ U|_{t=0} = U_0, \quad \dot{U}|_{t=0} = U_1, \text{ in } \Omega. \end{array} \right.$$

All notations have the same meaning as in [53].

These relations form the macroscopic problem solved in all numerical examples, with all effective tensors directly obtained from the unit-cell homogenization corresponding to a given damage level s .

4. Numerical experiments

In this section, we present numerical results obtained using the proposed fracture model to validate its effectiveness. Firstly, we establish the relationship between the components of the tensors $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{F}$ and the normalized damage variable s . Subsequently, we analyze the size effect and strain gradient effect in uniaxial tensile tests by computing the local macroscopic material response. Finally, we investigate the spalling test of alumina (AL23) in the last subsection.

4.1. Unit cell analysis for tensor $\mathbf{C}$, $\mathbf{D}$, $\mathbf{F}$

Based on the definitions of tensor $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{F}$ as presented in Eqs. (18), (26) and (29), it can be observed that each component is evaluated by integrating the first-order cell function in Y_s , which contains a square crack at its center. Therefore, these components are all functions with respect to the damage variable s . To account for the implicit dependence on s , we evaluate $C_{ijkl}(s)$, $D_{pqjmn}(s)$ and $F_{ijkl}(s)$ at 109 discrete points, where s takes values of $0.01^2, 0.02^2, \dots, 0.99^2, 0.991^2, 0.992^2, \dots, 0.999^2$. The numerical computations are performed using PMMA material, with mechanical parameters including a Young's modulus of $E = 3.16$ GPa, a Poisson ratio of $\nu = 0.32$ and a critical fracture energy of $G_c^0 = 350$ J/m².

Fig. 4 presents the variation curves of the independent components C_{ijkl} with respect to s . The results indicate that all nonzero components decrease as s increases, except for C_{1212} , which remains constant. Notably, C_{3333} exhibits a significant decrease, while C_{1111} decreases more slowly, owing to the crack surface being parallel to the Oy_1y_2 plane. Consequently, the stiffness reduction is more severe in the direction perpendicular to the micro-crack surface, while the stiffness reduction is less in the direction parallel to the micro-crack surface. For the initial state without damage ($s \rightarrow 0^+$), the modulus is computed as follows: $C_{1111}(0) = C_{3333}(0) = E(1 - \nu)/[(1 + \nu)(1 - 2\nu)]$, $C_{1122}(0) = C_{1133}(0) = E\nu/[(1 + \nu)(1 - 2\nu)]$ and $C_{1313}(0) = C_{1212}(0) = E/[2(1 + \nu)]$. For the fully damaged state ($s \rightarrow 1^-$), we approximate $C_{1111}(1^-)$, $C_{1122}(1^-)$, $C_{1313}(1^-)$ with a computable value of $1 - 1e^{-4}$ and approximate C_{1133} , C_{3333} with a small value ϵ , which should be set small enough to highlight the loss of tensile capacity when the material is completely broken.

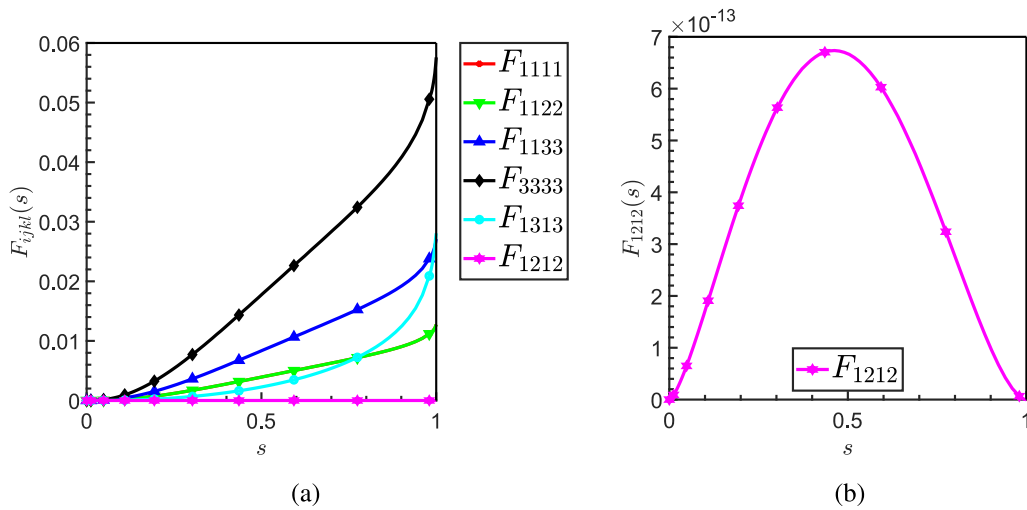


Fig. 5. $F_{ijkl} - s$ curve.

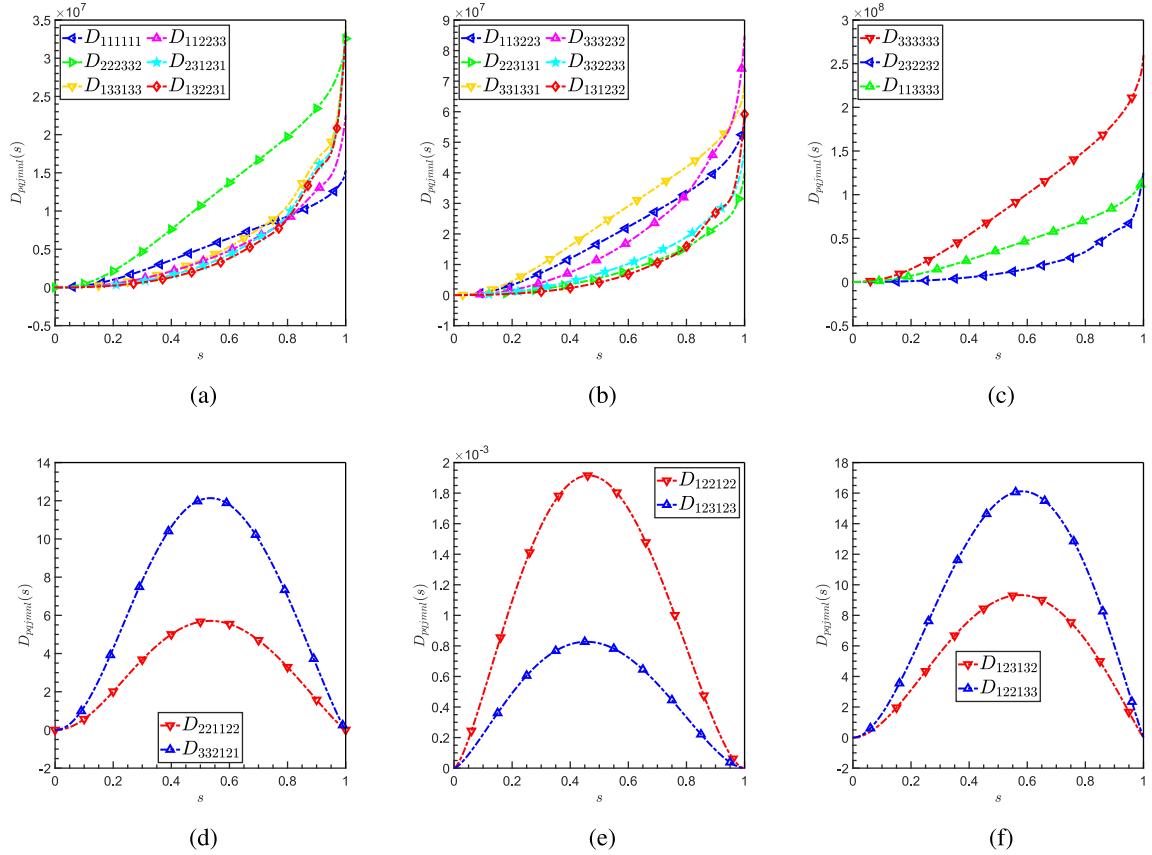
The variations of the independent components of tensor F with respect to s are presented in Fig. 5. It is evident that all nonzero independent components, except for F_{1212} , exhibit an increasing trend with respect to s . This trend indicates that the effect of strain rate becomes more significant as s increases. Notably, the component F_{1212} is negligible compared to other nonzero components in Fig. 5(b) and thus it is set to zero in subsequent numerical calculations. It is shown that F_{1212} first increases from 0 to a maximum value as s increases from 0 to approximately 0.5 and then decreases to 0 as s increases to 1.

Fig. 6 displays the independent non-zero components of tensor D as functions of the variable s . It can be observed from the first three subgraphs in Fig. 6 that all components of D increase monotonically as s increases from 0 to 1. However, from the last three subgraphs in Fig. 6, it can be inferred that each component first increases as s increases from 0 to a value near 0.5 and then decreases as s increases from 0.5 to 1, which is analogous to the behavior of the component F_{1212} of tensor F . Notably, all components in the last three subgraphs of Fig. 6, as well as the component F_{1212} , are related to the cell function $N^{12}(y)$ (refer to Fig. A.6). To prevent numerical instability, these components, which are at least six orders of magnitude smaller than the other components, are set to zero in subsequent sections. For an intact material with $s = 0$, the strain gradient has no effect since all components of tensor D are zero. This observation is consistent with the strain gradient theory for homogeneous materials in the existing literature [21,22].

In particular, the gradient-elastic tensor D develops coupling terms between the two in-plane directions. The non-monotonic evolution with respect to the damage variable observed in several components of D and F is directly linked to the cell function N^{12} associated with the two in-plane principal directions of the crack surface. Such behavior indicates that once the crack grows beyond a certain extent, the gradient effects in specific directions begin to diminish. This trend arises from the complex geometric interactions of the stress field along the three-dimensional crack front, reflecting the enriched anisotropy of the system and offering additional mechanistic insight into failure processes in three dimensions. Moreover, because the crack has a finite surface area, strain-gradient localization occurs simultaneously along the two in-plane directions, producing biaxial boundary layers. The displacement field may develop significant curvature both in the normal direction and within the crack plane, leading to substantially more complex bending modes in 3D. These characteristics explain why certain components (such as D_{221122} , F_{1212}) exhibit a rise-and-fall pattern rather than a monotonic trend: the competing influence of in-plane shear, curvature-induced deformation, and the progressive loss of stiffness normal to the crack surface leads to a peak in gradient sensitivity at intermediate damage levels. This non-monotonic behavior does not occur in a 2D setting, further demonstrating that the three-dimensional crack-surface geometry introduces additional deformation modes and stress-redistribution mechanisms that are absent in the planar model.

Remark 2. In the 2D setting of Rao et al. [47], under the plane-strain assumption, the mechanical response is constrained within a single plane, leading to 4, 12, and 4 independent nonzero components in the tensors C , D and F , respectively. In contrast, in the present three-dimensional formulation, the crack surface possesses two in-plane principal directions as well as a direction normal to the crack plane. Consequently, the damage-induced anisotropy becomes significantly richer in 3D. The tensors C , D and F contain 6, 21 and 5 independent nonzero components, respectively (see Appendix B).

Remark 3. The observed behavior of the gradient elastic moduli as $s \rightarrow 1$ can be understood from the fundamentally different response mechanisms of classical stiffness versus gradient stiffness when the load-bearing area diminishes. While the classical stiffness components associated with the transverse direction (e.g. C_{3333}) must vanish as the crack becomes through-thickness, the gradient elastic moduli $D_{pqijmnl}$ do not represent classical stiffness. Instead, they arise from the homogenized response of the strain-gradient fields in a highly heterogeneous microstructure. As s increases, the representative unit cell evolves toward a configuration consisting of two nearly separated ligaments, causing the strain-gradient fields to become increasingly concentrated near the crack

Fig. 6. $D_{pqjmmi} - s$ curve.

surfaces. This localization markedly enhances the effective energetic contribution of strain gradients, even as the classical transverse stiffness approaches zero.

4.2. Local macroscopic material response

In this subsection, we present the local macroscopic material response, which pertains to the mechanical behavior of a macroscopic material point, specifically the mechanical response of a Gaussian point in finite element simulations. The mechanical parameters outlined in Section 4.1 are utilized in our computations. The critical stress are denoted by Σ_c .

Firstly, we investigate a quasi-static uniaxial tensile test, wherein we exclude the influence of strain gradient and strain rate. The strain tensor's components, except for $\mathcal{E}_{33}$, are assumed to be zero, resulting in the tensile direction being perpendicular to the crack surface. This condition leads to fracture mode of Mode-I.

The curve of critical stress Σ_c versus $\epsilon(1/\sqrt{\epsilon})$ with initial damage $s = 0.01$ and $s = 0.25$ is shown in Fig. 7. It is observed that the critical stress decreases with an increase in ϵ for fixed damage s . Note that ϵ can be considered as the average spacing between micro-cracks. It is shown that Σ_c depends linearly on $1/\sqrt{\epsilon}$ from Fig. 7(b), which reveals the Hall-Petch relation in the literature. This is consistent with the analysis in the 2D case [25].

The effect of strain gradient is displayed in Fig. 8. With the increase of ϵ , the effect of strain gradient becomes more and more significant. All three curves almost coincide when ϵ is small enough, which means the effect of strain gradient is weakened if the homogenization of the material is strengthened [28]. On the other hand, we find that the larger $\mathcal{E}_{33}$ give the higher Σ_c by comparing the three curves. It can be explained that the stress concentration is weakened to a certain degree and the growth of micro-crack is retarded.

Next, we consider dynamic fracture and take the effect of strain rate into consideration. All components of the strain tensor and strain rate tensor are assumed to be zero except for $\mathcal{E}_{33}$ and $\dot{\mathcal{E}}_{33}$.

Fig. 9 displays the effect of strain rate on the damage-strain curve and stress-strain curve. It is observed that the damage variable (relative micro-crack area) sharply increases and stress sharply drops if the strain becomes large enough with a lower strain rate. So the material shows brittle fracture when the strain rate is low, for example, $\dot{\mathcal{E}}_{33} = 100 \text{ s}^{-1}$ as shown in Fig. 9. As the strain rate increases, the growth of micro-crack and drop of stress becomes gradually smoother, which means the ductility is more and more

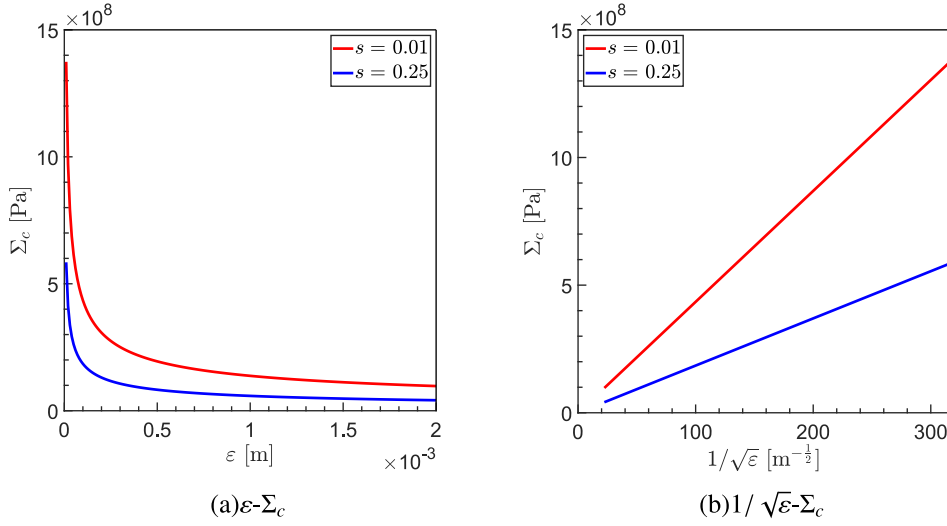


Fig. 7. The relationship between Σ_c and ε for different initial damages s .

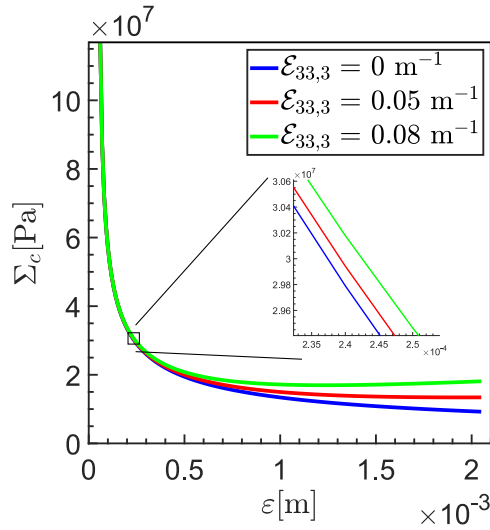


Fig. 8. Effect of strain gradient on the Σ_c - ε curve for initial damage $s = 0.25$.

significant. In addition, the tensile strength (critical stress) also increases with increasing strain rate. Actually, Σ_c is closely related to the kinetics of energy barriers, inertia effects and specimen size, which have been studied in former literature [35,36].

Fig. 10 displays the effects of microstructural size ε on the damage-strain and stress-strain curves with a constant strain rate of $\dot{\varepsilon}_{33} = 5 \times 10^3 \text{ s}^{-1}$ and an initial damage of $s = 0.04$. It is shown that s begins to increase at larger $\mathcal{E}_{33}$ for smaller ε from the damage-strain curve in Fig. 10, indicating that damage nucleation is sensitive to ε . We can also find that the critical stress Σ_c increases as ε decreases from the stress-strain curve in Fig. 10. Both curves predict an increasingly significant ductility with increasing ε . In fact, the heterogeneity of the material is strengthened for larger ε , which also strengthens the influence of strain rate on the energy release rate G by (36) intuitively and thus retards the propagation of micro-cracks. However, it gives a reverse result for large $\varepsilon = 0.005 \text{ m}$ if the strain rate is set to $\dot{\varepsilon}_{33} = 2 \times 10^4 \text{ s}^{-1}$ as shown in Fig. 11. This can also be explained by (36), which indicates that the energy release rate G decreases with increasing strain rate for uniaxial tensile tests. However, the effect of strain rate also depends on the coefficient ε^3 . Therefore, it is the result of the competition between the effect of microstructure ε and strain rate $\dot{\varepsilon}_{33}$.

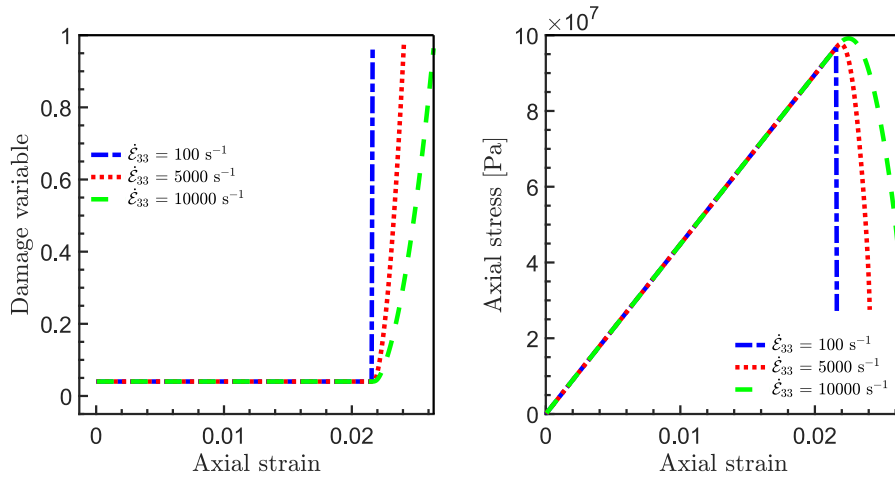


Fig. 9. Effect of strain rate $\dot{\epsilon}_{33}$ with $\epsilon = 1 \times 10^{-3}$ m and initial damage $s = 0.04$.

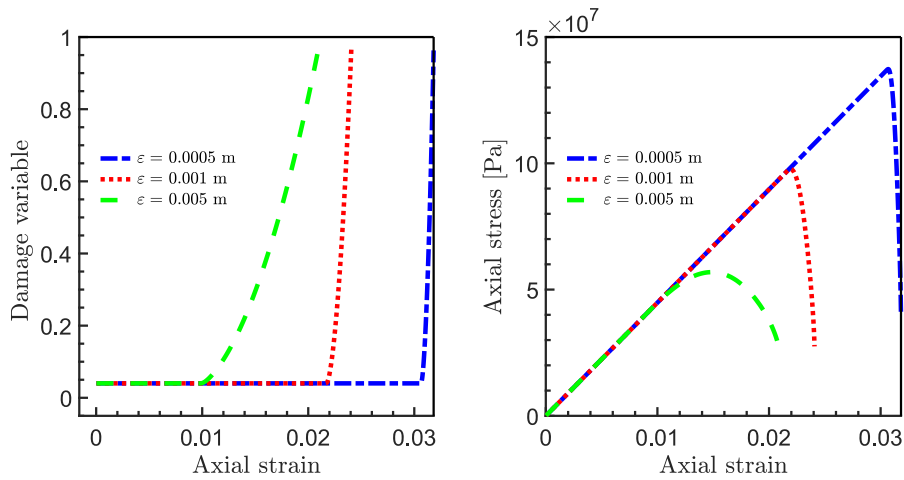


Fig. 10. Effect of ϵ with $\dot{\epsilon}_{33} = 5 \times 10^3$ s $^{-1}$ and initial damage $s = 0.04$.

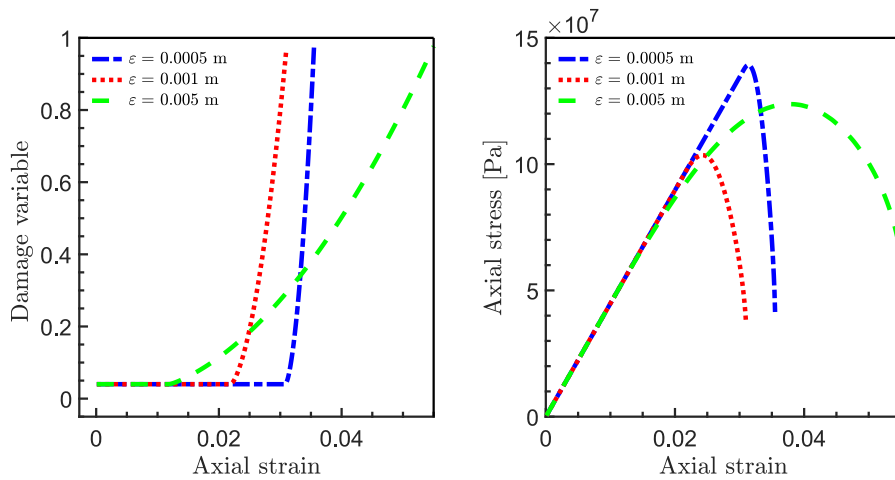


Fig. 11. Effect of ϵ with $\dot{\epsilon}_{33} = 2 \times 10^4$ s $^{-1}$ and initial damage $s = 0.04$.

Table 1
Physical and mechanical parameters of AL23 alumina.

E (GPa)	ν (-)	G_c^0 (J/m ²)	ρ (kg/m ³)
360	0.22	50.752	3890

4.3. Spalling fracture

Algorithm 1 Numerical algorithm for macroscopic dynamic damage evolution

```

1: Inputs:  $U^n, l^n, s^n = (l^n)^2$ .
2: Update macroscale configuration:  $\mathcal{E}_{ij}^n, \mathcal{E}_{ij,k}^n, \dot{\mathcal{E}}_{ij}^n = (\mathcal{E}_{ij}^n - \mathcal{E}_{ij}^{n-1})/\Delta t$ .
3: Update  $G$  at Gauss points.
4: for  $p = 1 : N$  do ▷  $N$ : number of Gauss points
5:   if ( $G_p > G_c^0$ ) then
6:     Solve  $G_p = G_c \left( \frac{\epsilon(l - l^n)}{2\Delta t} \right)$  by Newton's iteration to obtain  $l_p^{n+1}$ .
7:     Update  $l_p^{n+1} > l_p^n$  and force  $l_p^{n+1} = 1$  if  $l_p^{n+1} > 1$ .
8:   else
9:     Set  $l_p^{n+1} = l_p^n$  unchanged.
10:  end if
11: end for
12: Compute  $s^{n+1} = (l^{n+1})^2$  and Update  $C_{ijkl}(s^{n+1}), F_{ijkl}(s^{n+1}), D_{pqjmn}(s^{n+1})$ .
13: Solve equilibrium equation to obtain  $U^{n+1}$ .
14: Outputs:  $U^{n+1}, l^{n+1}$ .

```

In this subsection, we consider dynamic brittle fracture for macro-structures for spalling fracture. We solve the equilibrium equation (1) coupled with Eq. (37) to obtain the evolution of damage l (or $s = l^2$). For the calculation of $\dot{l}$, we use an explicit integral method to solve the following equation and update l in each time step.

$$G(l) = G_c \left(\frac{\epsilon \dot{l}}{2} \right). \quad (39)$$

As for spatial discretization, we adopt the finite element method with triangular meshes.

We use the Lagrangian P_2 element for displacement $\mathbf{u}$ and the Lagrangian P_0 element for the damage variable s , which means that the damage is constant in each triangular element. At each time step, we calculate the displacement variable and obtain the strain and strain gradient at the centroid of each triangular element. Then, we compute the evolution of the damage variable s by passing through the Gauss points in all triangular elements for the following calculations. The numerical algorithm governing macroscopic dynamic damage evolution is listed in Algorithm 1.

To explicate the effect of strain rate on macroscopic brittle fracture, we examine the shockless spalling test. Typically, spalling experiments entail the collision of two plate materials, wherein one plate (the flyer) is accelerated to an initial velocity and subsequently impacts the other plate (the target). Upon impact, a shockwave is generated from the impact surface, which subsequently transforms into two rarefaction waves after reflection at the free surfaces. The interaction between these waves generates a region of tension within the target plate, which can culminate in spall failure within the interior of the target plate if the tensile stress surpasses the fracture strength of the material [38].

To facilitate a comparison between the results of the GEPI spalling tests in [40] and numerical simulations, we utilize a cuboid configuration measuring 10 mm in height and 2.5 mm in length and width, as depicted in Fig. 12. The target material selected for the simulation is AL23 alumina, whose material parameters are presented in Table 1. Symmetry conditions are imposed on both internal faces, while a zero radial displacement boundary condition ($U_r = 0$) is applied to the two external faces to ensure a uniaxial strain state during the computation. The upper face is set to a free boundary, while a velocity boundary condition is applied to the lower face. Point O on the rear free-surface denotes the node where the free surface velocity profile is captured. The mesh size adopted for the simulation is $h \approx 0.3$ mm. Four distinct velocity-time loading curves are tested, as illustrated in Fig. 13, corresponding to the four loading conditions utilized in the experiments of Zinszner et al. [40].

The damage zone evolution under wave 1 impact loading is simulated using the three discretization schemes depicted in Fig. 14. As shown in Fig. 15, the predicted damage profiles exhibit convergence between mesh 2 and mesh 3 with progressive refinement, quantitatively validating the attenuation of mesh dependency with enhanced spatial resolution.

Fig. 16 displays the fracture zone with an initial damage of $s_0 = 0.01$ and a microstructure parameter of $\epsilon = 1$ mm, induced by the application of stress waves as illustrated in Fig. 13. The results demonstrate a strong dependence of the size of the damaged zone on the strain rate. Furthermore, it is observed that the maximum distance between the damaged zone and the free surface increases with increasing strain rate, while the minimum distance between the damaged zone and the free surface decreases with increasing strain rate. The damaged zone becomes denser and larger as the strain rate increases. Notably, incomplete damage is observed in the distal damage zone for large strain rates.

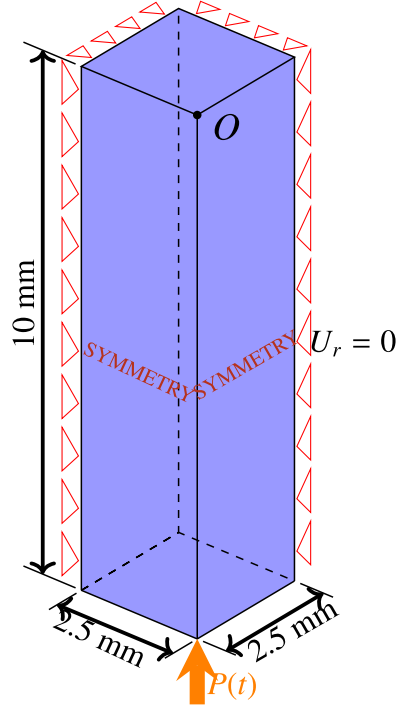


Fig. 12. Geometric configuration and boundary conditions of spallation numerical test.

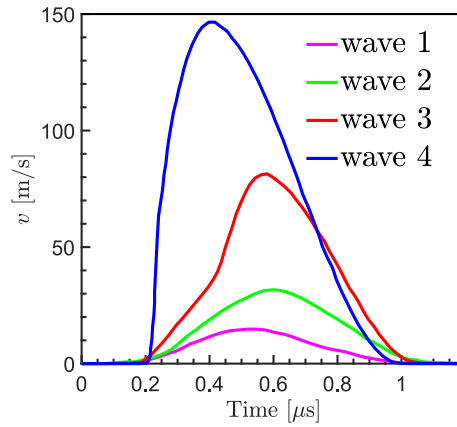


Fig. 13. Loading velocity history for inducing stress waves in the spall simulations.

Fig. 17 displays the fracture zone with different initial damage values of $s_0 = 0.01$, $s_0 = 0.03$ and $s_0 = 0.04$, along with a microstructure parameter of $\varepsilon = 1$ mm, induced by wave 1 in Fig. 13. The label variable s_0 for the colorbar denotes the initial damage. The results demonstrate that the calculated fracture zone prediction for $s_0 = 0.01$ is in good agreement with the experimental results, especially with the damaged zone located approximately 2.1 mm away from the free surface. The maximum distance between the damaged zone and the free surface is approximately 4.4 mm for all three numerical tests, which is consistent with the experimental outcomes. The final fracture zone is observed to be larger for larger initial damage values and incomplete damage is observed in the distal damage zone for larger values of s_0 . The dominant components of $\mathbf{D}$ and $\mathbf{F}$ increase with the increase of s , as illustrated in Figs. 5 and 6, leading to a more significant influence of strain gradient and strain rate and a retardation in the evolution of damage, ultimately culminating in incomplete damage in the distal damage zone.

In spalling experiments, determining the ultimate tensile stress σ_{spalling} is generally challenging. However, it can be computed using the classical acoustic approximation formulated by Novikov et al. [54] as follows:

$$\sigma_{\text{spalling}} = \frac{1}{2} \rho C_L \Delta V_{\text{pb}},$$

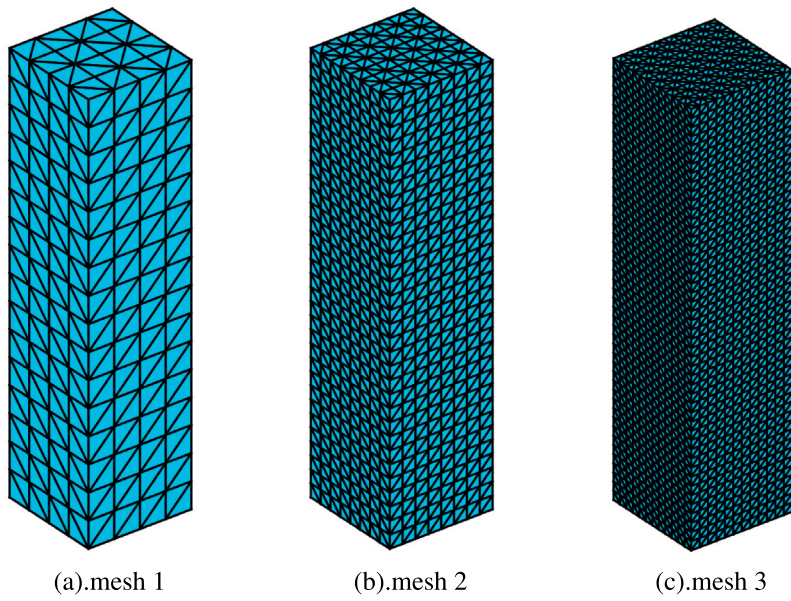


Fig. 14. Three different finite element meshes.

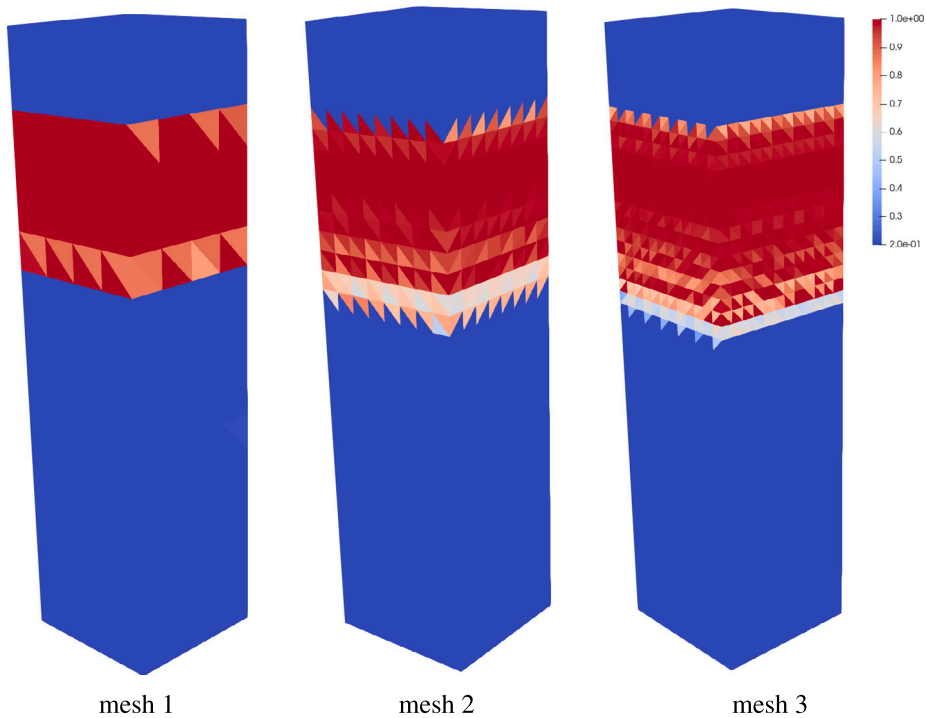


Fig. 15. Fracture zone obtained by applying wave 1 with $s_0 = 0.02$ and $\varepsilon = 1$ mm for three meshes.

here, ρ denotes the density, C_L represents the longitudinal wave speed and ΔV_{pb} is the pullback velocity. The pullback velocity is defined as the difference between the maximum velocity measured on the rear free-surface and the first rebound of velocity, which is indicative of the tensile damage within the core of the tested sample. Therefore, the rear free surface velocity profile is one of the most frequently measured quantities in experiments.

In numerical simulations, the velocity histories of the central point O at the rear surface are recorded and compared to experimental results. Fig. 18 shows a comparison between experimental and numerical results for free surface velocities, where

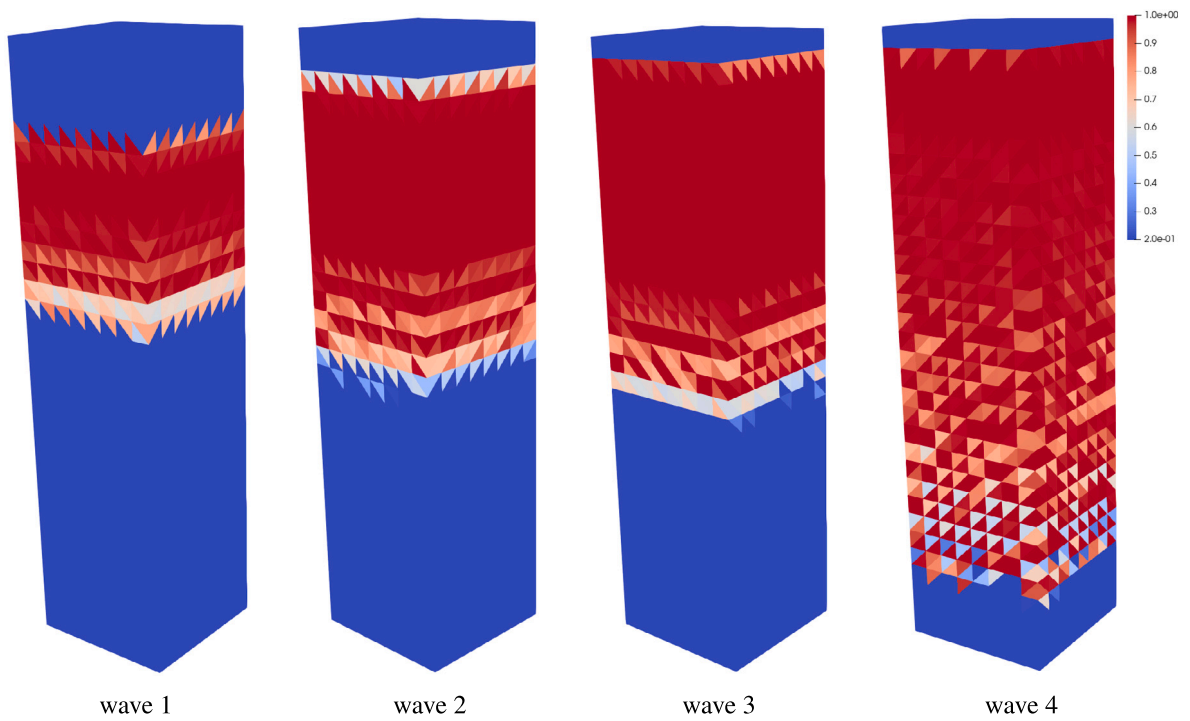


Fig. 16. Fracture zone obtained by applying different stress waves with $s_0 = 0.04$ and $\epsilon = 1$ mm.

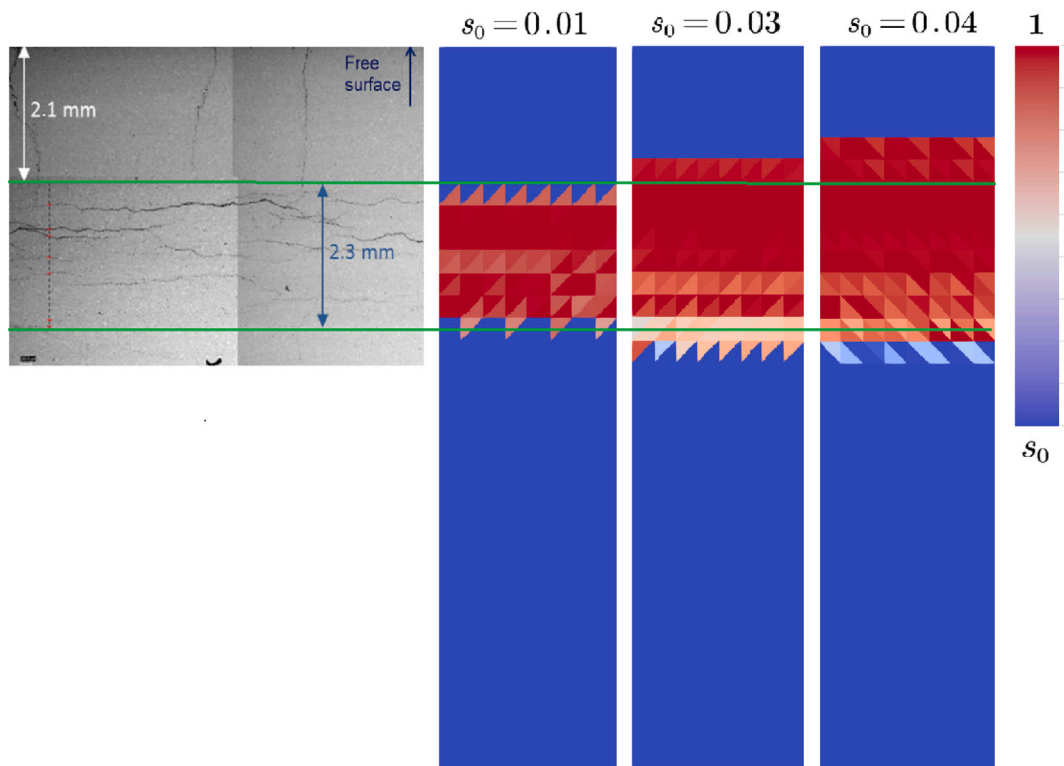


Fig. 17. Comparing the fracture zone in the spall experiment [40] and numerical results with different initial damage.

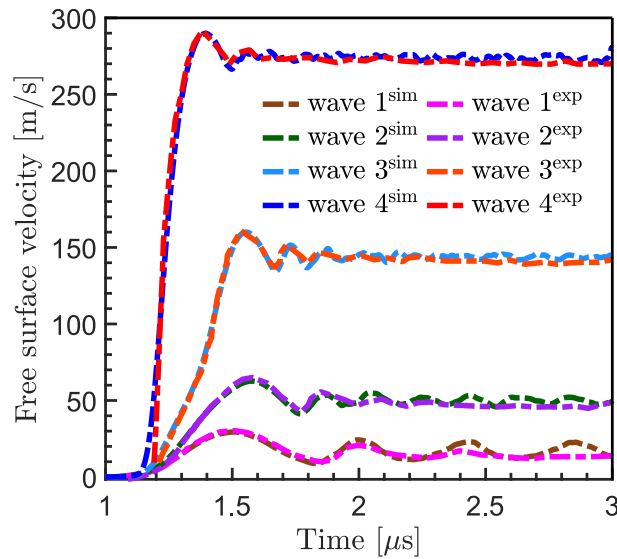


Fig. 18. Comparison of numerical and experimental results for free surface velocity curve.

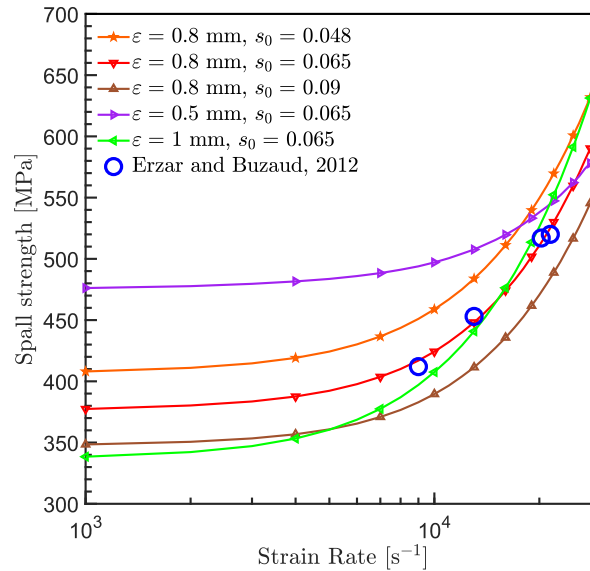


Fig. 19. Effect of strain-rate on tensile strength: Comparison of calculated results with experimental measurement.

the velocity waves displayed in Fig. 13 were applied. The dashed lines with the superscript “exp” represent the experimental results obtained in [40], while the dashed lines with superscript “sim” represent the simulation results in the present work.

Due to the lack of experimental data for the AL23 alumina sample, such as initial damage s_0 and microstructure parameter ϵ , we need to calibrate these parameters to obtain ideal agreement with the experimental results. We use different initial damage s_0 and microstructure parameter ϵ for each group to carry out numerical experiments. For the results displayed in Fig. 18, we adopt microstructure parameter $\epsilon = 1.2$ mm and 0.8 mm for waves 1 and 2, respectively and $\epsilon = 0.5$ mm for waves 3 and 4. We also adopt initial damage $s_0 = 0.023$ and 0.063 for waves 1 and 4, respectively and $s_0 = 0.04$ for waves 2 and 3. It is observed that the velocities are in good agreement with the experimental results in the early stage, especially when $t < 2$ μ s, which demonstrates the validity of the model. However, the free surface velocities are slightly higher than those in the experiments, which might be due to the fact that our damage model is established only for brittle fracture and does not consider plastic damage and the effect of thermal energy. It should be improved to a more accurate model in future work.

To validate our model, we consider the spalling fracture experiment carried out by Erzár and Buzaud [39] and give the numerical results in Fig. 19. It is observed that the calculated numerical results agreed well with the experimental results if microstructure parameter $\epsilon = 0.8$ mm and initial damage $s_0 = 0.065$ are adopted. For a fixed initial damage $s_0 = 0.065$, the strain rate sensitivity of

the spall strength (slope of the curve at the same strain rate) increases with the increase of ϵ ; while for a fixed microstructure size $\epsilon = 0.8$ mm, the strain rate sensitivity of the spall strength varies insignificantly with the change of s_0 .

Remark 4. In the present study, the size of the selected representative volume element (RVE) is taken as $\epsilon = 500 \sim 1200$ μm , which corresponds to the average microcrack spacing. The initial damage parameter s_0 , being a dimensionless variable, represents the initial area fraction of the crack surface and corresponds to microcracks with an effective size of $50 \sim 120$ μm . For these parameters, our model qualitatively captures the underlying physical laws, and quantitative comparisons were achieved through parameter calibration.

5. Conclusion

In this paper, a unified approach is established to simultaneously account for the influences of strain gradient, strain rate and microstructure size on brittle fracture in three dimensions. Since macro fracture is the result of micro-crack propagation following the Griffith criterion, the brittle fracture model, previously established by two-scale asymptotic analysis in two dimensions [25,47], is extended to the three-dimensional case. In contrast to using the normalized micro-crack length d in two-dimensional problems, the normalized micro-crack area ($s = l^2$) is viewed as a measure of damage for a cell, with $s = 1$ representing the state of complete fracture. The strain-related term, strain gradient-related term and strain rate-related term are all incorporated in the local energy release rate of micro-crack propagation. By introducing the critical energy release rate, a speed-dependent function $G_c(v)$, we establish the macroscopic damage evolution equation based on the Griffith criterion. The form of the macroscopic damage evolution equation is consistent with the 2D case. The coefficients associated with strain, strain gradient and strain rate are expressed as derivatives of tensor functions **C**, **D** and **F** with respect to the normalized micro-crack area s .

Importantly, although the mathematical structure of the evolution equation resembles the 2D formulation, the present work provides a non-trivial extension. The transition from a line defect to a finite crack surface leads to fundamentally different 3D cell problems, additional independent components in the effective tensors, and new anisotropic degradation and gradient-amplification mechanisms. These features produce quantitative and qualitative behaviors that cannot be captured by a 2D model, thereby demonstrating the intrinsically three-dimensional nature of the proposed formulation.

Despite the physical assumptions are simple, the model exhibits remarkable predictive capabilities. The strain gradient and strain rate terms both lead to lower calculated local energy release rates. Therefore, an increase in strain gradient or strain rate results in enhanced fracture strength, fracture toughness and ductility. Both strain gradient and strain rate effects are significant for larger microstructural sizes ϵ (micro-crack separation). These findings are validated by numerical examples in the local analysis of mechanical response. The numerical spalling test carried out in [47] is also tested in 3D using finite element simulations. The simulation of impact-induced spalling fracture of AL23 alumina material effectively reproduces the fracture zones measured in experiments. Additionally, the calculated fracture strength of AL23 alumina material as a function of strain rate by the model is well-consistent with experimental measurements.

Overall, the present 3D asymptotic-homogenization framework provides a theoretically consistent and physically interpretable route to bridge microstructural crack evolution with macroscopic dynamic fracture behavior. Compared with the 2D counterpart, the proposed 3D model captures additional deformation modes, enriched gradient effects, and crack-surface-induced micro-inertia, offering new insight into the multiscale mechanisms governing dynamic brittle fracture. The modeling approach can be used as a paradigm for theoretically linking influence factors of macro-mechanical responses to microstructure evolutions in an ab-initio manner.

CRediT authorship contribution statement

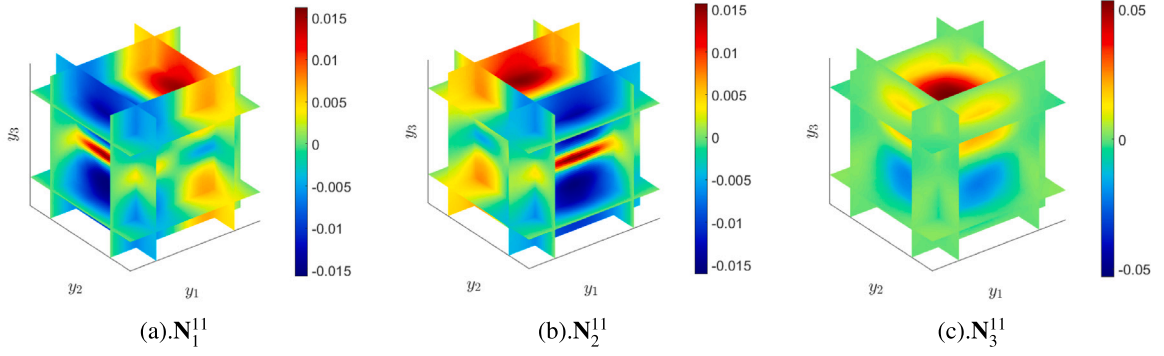
Yipeng Rao: Writing – original draft, Visualization, Software, Formal analysis, Conceptualization. **Quanzhang Li:** Writing – review & editing, Validation. **Zhiqiang Yang:** Writing – review & editing, Validation. **Meizhen Xiang:** Writing – review & editing, Validation, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Fig. A.1. Cell function $\mathbf{N}^{11}(\mathbf{y})$.

Appendix A. Symmetry of the cell function $\mathbf{N}^{ij}(\mathbf{y})$

Fig. 2 illustrates that the unit cell containing a squared crack in the center possesses three planes of symmetry, namely $y_1 = 0.5$, $y_2 = 0.5$ and $y_3 = 0.5$, among the planes parallel to the coordinate plane, while the other two planes of symmetry are $y_1 = y_2$ and $y_1 = 1 - y_2$. The relationship between different cell functions can be derived based on the symmetries of the cell's geometric configuration. These characteristics, combined with the symmetry of a_{ijkl} , can effectively reduce the number of independent components of tensors $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{F}$, thereby facilitating numerical calculations and resulting in more convenient and compact expressions.

We consider the weak form of (15). Multiplying the first equation of (15) by v_i and integrating over Y_s , we obtain:

$$\int_{Y_s} a_{ijpq} e_{yij}(\mathbf{v}) + a_{ijmn} e_{ymn}(\mathbf{N}^{pq}) e_{yij}(\mathbf{v}) d\mathbf{y} = 0, \quad (\text{A.1})$$

here, v_i is the i th component of vector valued function $\mathbf{v} \in D(Y_s)$.

Firstly, we consider the case where $p = 1$ and $q = 1$. We can prove the following symmetry and anti-symmetry of $\mathbf{N}^{11}(y_1, y_2, y_3)$ (see Fig. A.1):

$$\begin{aligned} \mathbf{N}_1^{11}(y_1, y_2, y_3) &= -\mathbf{N}_1^{11}(1 - y_1, y_2, y_3) = \mathbf{N}_1^{11}(y_1, 1 - y_2, y_3) = \mathbf{N}_1^{11}(y_1, y_2, 1 - y_3); \\ \mathbf{N}_2^{11}(y_1, y_2, y_3) &= \mathbf{N}_2^{11}(1 - y_1, y_2, y_3) = -\mathbf{N}_2^{11}(y_1, 1 - y_2, y_3) = \mathbf{N}_2^{11}(y_1, y_2, 1 - y_3); \\ \mathbf{N}_3^{11}(y_1, y_2, y_3) &= \mathbf{N}_3^{11}(1 - y_1, y_2, y_3) = \mathbf{N}_3^{11}(y_1, 1 - y_2, y_3) = -\mathbf{N}_3^{11}(y_1, y_2, 1 - y_3). \end{aligned} \quad (\text{A.2})$$

We define a vector-valued function $\tilde{\mathbf{N}}^{11}(\mathbf{y}) = (-\mathbf{N}_1^{11}(1 - y_1, y_2, y_3), \mathbf{N}_2^{11}(1 - y_1, y_2, y_3), \mathbf{N}_3^{11}(1 - y_1, y_2, y_3))$. For any given function $\mathbf{v}(\mathbf{y})$, we consider a vector-valued function $\tilde{\mathbf{v}}(\mathbf{y}) = (-v_1(1 - y_1, y_2, y_3), v_2(1 - y_1, y_2, y_3), v_3(1 - y_1, y_2, y_3))$.

Substituting $\tilde{\mathbf{N}}^{11}(\mathbf{y})$, $\tilde{\mathbf{v}}(\mathbf{y})$ into (A.1), the weak form of $\tilde{\mathbf{N}}^{11}(\mathbf{y})$ is:

$$\int_{Y_s} a_{ij11} e_{yij}(\tilde{\mathbf{v}}) + a_{ijmn} e_{ymn}(\tilde{\mathbf{N}}^{11}) e_{yij}(\tilde{\mathbf{v}}) d\mathbf{y} = 0, \quad (\text{A.3})$$

which has the same form as the case $p = 1, q = 1$ in (A.1). By the arbitrariness of $\mathbf{v}(\mathbf{y})$ and the existence and uniqueness of the variational problem, we have $\mathbf{N}^{11}(\mathbf{y}) = \tilde{\mathbf{N}}^{11}(\mathbf{y})$. The remaining equations in (A.2) can be proven similarly.

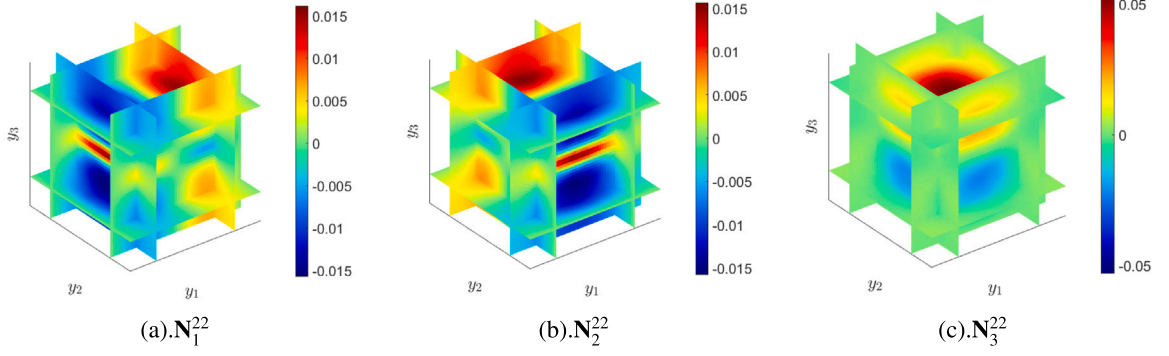
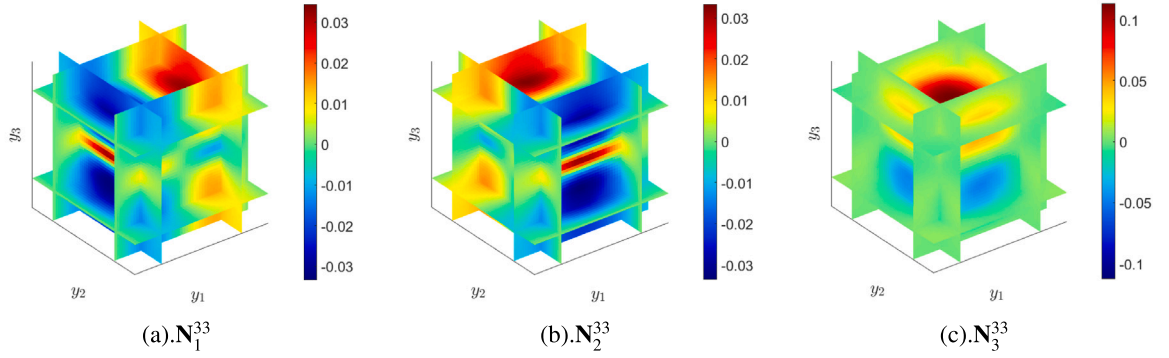
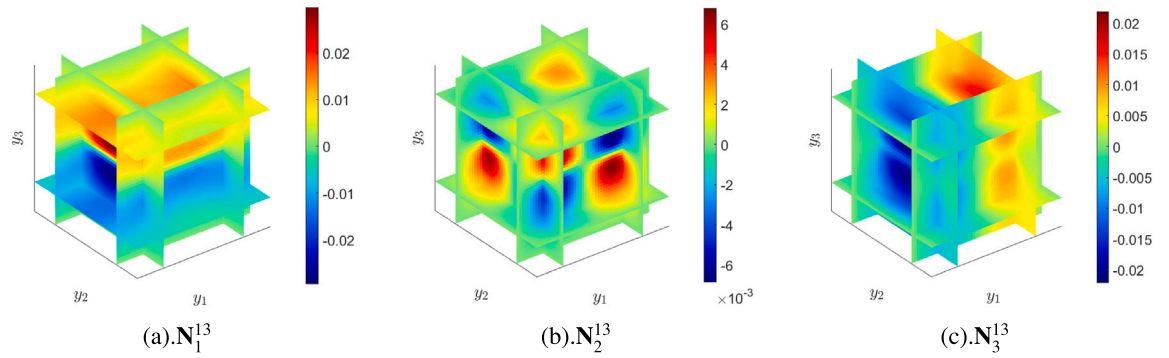
We shall now demonstrate the symmetry between $\mathbf{N}^{11}(\mathbf{y})$ and $\mathbf{N}^{22}(\mathbf{y})$, which arises from the geometric symmetry of the planes defined by $y_1 = y_2$ and $y_1 = 1 - y_2$.

$$\begin{aligned} \mathbf{N}_1^{11}(y_1, y_2, y_3) &= \mathbf{N}_2^{22}(y_2, y_1, y_3) = \mathbf{N}_1^{22}(y_1, y_2, y_3); \\ \mathbf{N}_2^{11}(y_1, y_2, y_3) &= \mathbf{N}_1^{22}(y_2, y_1, y_3) = \mathbf{N}_2^{22}(y_1, y_2, y_3); \\ \mathbf{N}_3^{11}(y_1, y_2, y_3) &= \mathbf{N}_3^{22}(y_2, y_1, y_3) = \mathbf{N}_3^{22}(y_1, y_2, y_3). \end{aligned} \quad (\text{A.4})$$

We introduce a vector-valued function $\check{\mathbf{N}}^{22}(y_1, y_2, y_3) = (\mathbf{N}_2^{22}(y_2, y_1, y_3), \mathbf{N}_1^{22}(y_2, y_1, y_3), \mathbf{N}_3^{22}(y_2, y_1, y_3))$ and define a vector-valued function $\check{\mathbf{v}}(y_1, y_2, y_3) = (v_2(y_2, y_1, y_3), v_1(y_2, y_1, y_3), v_3(y_2, y_1, y_3))$. By substituting $\check{\mathbf{N}}^{22}(\mathbf{y})$, $\check{\mathbf{v}}(\mathbf{y})$ into (A.1), along with the conditions $a_{1111} = a_{2222}$ and $a_{1133} = a_{2233}$, we can express the weak form of $\check{\mathbf{N}}^{22}(\mathbf{y})$ as follows:

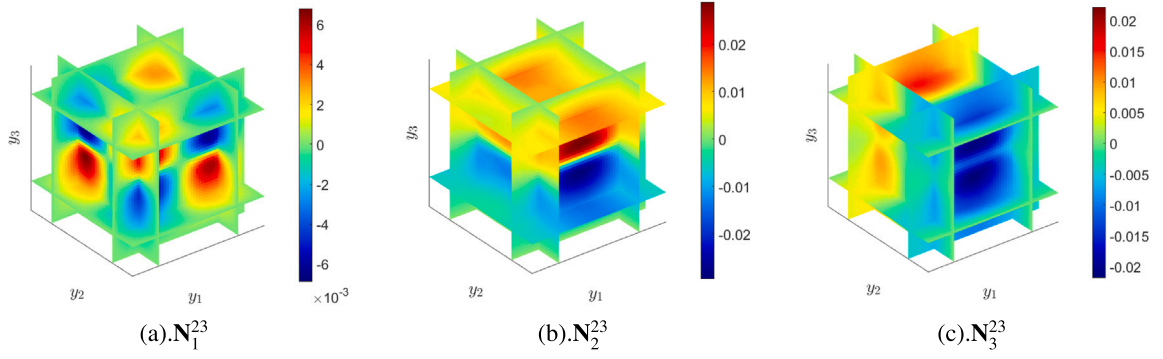
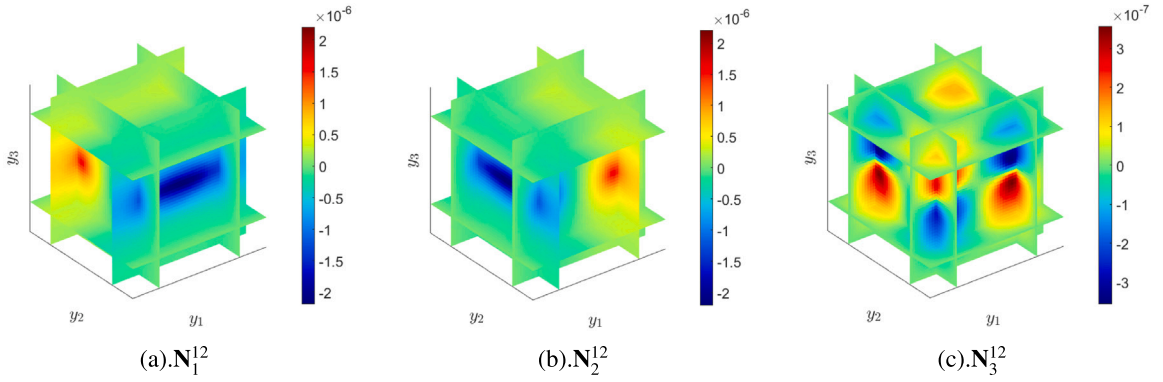
$$\int_{Y_s} a_{ij11} e_{yij}(\check{\mathbf{v}}) + a_{ijmn} e_{ymn}(\check{\mathbf{N}}^{22}) e_{yij}(\check{\mathbf{v}}) d\mathbf{y} = 0. \quad (\text{A.5})$$

Note that (A.5) has the same form as the case $p = 1, q = 1$ in (A.1), so we have $\check{\mathbf{N}}^{22}(\mathbf{y}) = \mathbf{N}^{11}(\mathbf{y})$. By directly comparing the weak

Fig. A.2. Cell function $N^{22}(\mathbf{y})$.Fig. A.3. Cell function $N^{33}(\mathbf{y})$.Fig. A.4. Cell function $N^{13}(\mathbf{y})$.

forms of $N^{11}(\mathbf{y})$ and $N^{22}(\mathbf{y})$, we obtain $N^{11}(\mathbf{y}) = N^{22}(\mathbf{y})$. Therefore, we have confirmed (A.4). This can also be verified by comparing the plots in Figs. A.1 and A.2.

To reflect the symmetry and antisymmetry of the cell functions more intuitively, we provide the numerical results of $N^{33}(\mathbf{y})$, $N^{13}(\mathbf{y})$, $N^{23}(\mathbf{y})$ and $N^{12}(\mathbf{y})$. We then present the relationships between different cell functions without formal proof.

Fig. A.5. Cell function $N^{23}(\mathbf{y})$.Fig. A.6. Cell function $N^{12}(\mathbf{y})$.

The relationships between different cell functions are listed as follows:

$$\begin{cases}
 N_1^{11}(y_1, y_2, y_3) = N_2^{22}(y_2, y_1, y_3) = N_1^{22}(y_1, y_2, y_3); \\
 N_2^{11}(y_1, y_2, y_3) = N_1^{22}(y_2, y_1, y_3) = N_2^{22}(y_1, y_2, y_3); \\
 N_3^{11}(y_1, y_2, y_3) = N_3^{22}(y_2, y_1, y_3) = N_3^{22}(y_1, y_2, y_3). \\
 N_1^{33}(y_1, y_2, y_3) = N_2^{33}(y_2, y_1, y_3). \\
 \begin{cases}
 N_2^{13}(y_1, y_2, y_3) = N_1^{23}(y_2, y_1, y_3); \\
 N_1^{13}(y_1, y_2, y_3) = N_2^{23}(y_2, y_1, y_3); \\
 N_3^{13}(y_1, y_2, y_3) = N_3^{23}(y_2, y_1, y_3).
 \end{cases} \\
 N_1^{12}(y_1, y_2, y_3) = N_2^{12}(y_2, y_1, y_3).
 \end{cases} \quad (\text{A.6})$$

Based on the above equations, there are only ten independent cell functions. We present the results of these independent cell functions below and the symmetry and anti-symmetry of all cell functions can be obtained using (A.6).

$$\begin{cases}
 N_1^{11}(y_1, y_2, y_3) = -N_1^{11}(1 - y_1, y_2, y_3) = N_1^{11}(y_1, 1 - y_2, y_3) = N_1^{11}(y_1, y_2, 1 - y_3); \\
 N_2^{11}(y_1, y_2, y_3) = N_2^{11}(1 - y_1, y_2, y_3) = -N_2^{11}(y_1, 1 - y_2, y_3) = N_2^{11}(y_1, y_2, 1 - y_3); \\
 N_3^{11}(y_1, y_2, y_3) = N_3^{11}(1 - y_1, y_2, y_3) = N_3^{11}(y_1, 1 - y_2, y_3) = -N_3^{11}(y_1, y_2, 1 - y_3). \\
 \begin{cases}
 N_1^{33}(y_1, y_2, y_3) = -N_1^{33}(1 - y_1, y_2, y_3) = N_1^{33}(y_1, 1 - y_2, y_3) = N_1^{33}(y_1, y_2, 1 - y_3); \\
 N_3^{33}(y_1, y_2, y_3) = N_3^{33}(1 - y_1, y_2, y_3) = N_3^{33}(y_1, 1 - y_2, y_3) = -N_3^{33}(y_1, y_2, 1 - y_3).
 \end{cases} \\
 \begin{cases}
 N_1^{12}(y_1, y_2, y_3) = N_1^{12}(1 - y_1, y_2, y_3) = -N_1^{12}(y_1, 1 - y_2, y_3) = N_1^{12}(y_1, y_2, 1 - y_3); \\
 N_3^{12}(y_1, y_2, y_3) = -N_3^{12}(1 - y_1, y_2, y_3) = -N_3^{12}(y_1, 1 - y_2, y_3) = -N_3^{12}(y_1, y_2, 1 - y_3).
 \end{cases} \\
 \begin{cases}
 N_1^{13}(y_1, y_2, y_3) = N_1^{13}(1 - y_1, y_2, y_3) = N_1^{13}(y_1, 1 - y_2, y_3) = -N_1^{13}(y_1, y_2, 1 - y_3); \\
 N_2^{13}(y_1, y_2, y_3) = -N_2^{13}(1 - y_1, y_2, y_3) = -N_2^{13}(y_1, 1 - y_2, y_3) = -N_2^{13}(y_1, y_2, 1 - y_3); \\
 N_3^{13}(y_1, y_2, y_3) = -N_3^{13}(1 - y_1, y_2, y_3) = N_3^{13}(y_1, 1 - y_2, y_3) = N_3^{13}(y_1, y_2, 1 - y_3).
 \end{cases}
 \end{cases}$$

Table A.1

Odd and even symmetries of unit cell functions.

	N_1^{11}	N_2^{11}	N_3^{11}	N_1^{22}	N_2^{22}	N_3^{22}	N_1^{33}	N_2^{33}	N_3^{33}
y_1	1	0	0	1	0	0	1	0	0
y_2	0	1	0	0	1	0	0	1	0
y_3	0	0	1	0	0	1	0	0	1
	N_1^{13}	N_2^{13}	N_3^{13}	N_1^{23}	N_2^{23}	N_3^{23}	N_1^{12}	N_2^{12}	N_3^{12}
y_1	0	1	1	1	0	0	0	1	1
y_2	0	1	0	1	0	1	1	0	1
y_3	1	1	0	1	1	0	0	0	1

We now proceed to prove the following equality, which is used to demonstrate (24), based on the aforementioned symmetries:

$$\int_{Y_\epsilon} N_{i,j}^{pq} N_k^{mn} dx = 0.$$

For the function N_k^{mn} , if $N_k^{mn}(y_1, y_2, y_3) = N_k^{mn}(1 - y_1, y_2, y_3)$, we refer to it as *even symmetric* with respect to y_1 (denoted as “0” in Table A.1); if $N_k^{mn}(y_1, y_2, y_3) = -N_k^{mn}(1 - y_1, y_2, y_3)$, we refer to it as *odd symmetric* with respect to y_1 (denoted as “1” in Table A.1). Similarly, the definitions of even and odd symmetry apply with respect to y_2 and y_3 . The odd and even symmetries of the unit cell functions are summarized in Table A.1. It can be readily shown that each partial derivative of N_k^{mn} with respect to y_i alters the parity of symmetry in the corresponding component.

To prove that the integral of $N_{i,j}^{pq} N_k^{mn}$ over Y_ϵ is zero, it suffices to show that $N_{i,j}^{pq} N_k^{mn}$ is antisymmetric with respect to at least one of the variables y_1, y_2, y_3 . From this, we obtain the following five parity cases for N_i^{pq} and N_k^{mn} . We demonstrate that for each case, the integral of $N_{i,j}^{pq} N_k^{mn}$ over Y_ϵ is zero.

- **Case (1,0,0) - (1,0,0)**

If differentiated with respect to y_1 , the parity of $N_{i,j}^{pq} N_k^{mn}$ becomes (1,0,0), so its integral over Y_ϵ is zero. If differentiated with respect to y_2 (similarly for y_3), the parity becomes (0,1,0), so the integral over Y_ϵ is zero.

- **Case (0,1,0) - (1,0,0)**

If differentiated with respect to y_1 (similarly for y_2), the parity becomes (0,1,0), so the integral over Y_ϵ is zero. If differentiated with respect to y_3 , the parity becomes (1,1,1), so the integral over Y_ϵ is zero.

- **Case (1,0,0) - (1,1,1)**

If differentiated with respect to y_1 , the parity becomes (1,1,1), so the integral over Y_ϵ is zero. If differentiated with respect to y_2 (similarly for y_3), the parity becomes (0,0,1), so the integral over Y_ϵ is zero.

- **Case (1,1,1) - (1,0,0)**

If differentiated with respect to y_1 , the parity becomes (1,1,1), so the integral over Y_ϵ is zero. If differentiated with respect to y_2 (similarly for y_3), the parity becomes (0,0,1), so the integral over Y_ϵ is zero.

- **Case (1,1,1) - (1,1,1)**

If differentiated with respect to y_1 (similarly for y_2, y_3), the parity becomes (1,0,0), so the integral over Y_ϵ is zero.

Appendix B. Independent components of tensor C, D, F

From the definitions of C, D and F in (18), (26) and (29), we can derive the following symmetries:

$$C_{ijkl} = C_{jikl} = C_{klij} \quad D_{pqjmn} = D_{qpjmn} = D_{mnlpq} \quad F_{ijkl} = F_{jikl} = F_{klij}. \quad (\text{B.1})$$

We now consider the symmetry of cell function obtained in Appendix A. Specifically, the symmetry of the planes $y_1 = y_2$ and $y_1 = 1 - y_2$ leads to the symmetry with the subscript $i = 1, 2$. This symmetry yields the relations $C_{1111} = C_{2222}$, $C_{1133} = C_{2233}$ and $C_{1313} = C_{2323}$ for tensor C and $F_{1111} = F_{2222} = F_{1122}$, $F_{1133} = F_{2233}$ and $F_{1313} = F_{2323}$ for tensor F. Finally, we have determined six independent components of tensor C and five independent components of tensor F. The matrix representations of tensor C and tensor F are presented in Figs. B.1 and B.2, respectively. It is worth noting that the same color, except black, is utilized to denote equal components. For instance, in Fig. B.1, we use red, orange and blue to indicate the equality of C_{1111} with C_{2222} , C_{1133} with C_{2233} and C_{2323} with C_{1313} , respectively. Meanwhile, the components C_{3333} , C_{1122} and C_{1212} are unique and hence marked in black.

The tensor D comprises $3^6 = 729$ components. However, the number of independent components reduces to 51 due to the symmetry and anti-symmetry of the cell function as well as the second equation in (B.1). The detailed analysis would not be given here and we only provide the matrix representation of D in Fig. B.3. The matrix shows four diagonal blocks, with the first two 5×5 blocks being identical. Within each block, we use different colors to denote unequal components, except for the black color, which represents a unique component. For instance, in the fourth 3×3 block in the diagonal, we observe that $D_{231123} = D_{123132}$ and $D_{231231} = D_{132132}$, the components D_{123123} and D_{231132} are unique and hence marked in black.

$$\left(\begin{array}{ccc|c} C_{1111} & C_{1122} & C_{1133} & \\ & C_{2222} & C_{2233} & \\ & & C_{3333} & \\ \hline & & & C_{1212} \\ & & & C_{2323} \\ & & & C_{1313} \end{array} \right)$$

Fig. B.1. The structure of tensor C.

$$\left(\begin{array}{ccc|c} F_{1111} & F_{1122} & F_{1133} & \\ & F_{2222} & F_{2233} & \\ & & F_{3333} & \\ \hline & & & F_{1212} \\ & & & F_{2323} \\ & & & F_{1313} \end{array} \right)$$

Fig. B.2. The structure of tensor F.

$$\left(\begin{array}{cccc|cccc|cccc|cccc} D_{111111} & D_{111221} & D_{111122} & D_{111331} & D_{111133} & & & & & & & & & & & & \\ & D_{221221} & D_{221122} & D_{221331} & D_{221133} & & & & & & & & & & & & & \\ & & D_{122122} & D_{122331} & D_{122133} & & & & & & & & & & & & & \\ & & & D_{331331} & D_{331133} & & & & & & & & & & & & & \\ & & & & D_{133133} & & & & & & & & & & & & & \\ \hline & & & & & D_{222222} & D_{222112} & D_{222121} & D_{222332} & D_{222233} & & & & & & & & \\ & & & & & & D_{112112} & D_{112121} & D_{112332} & D_{112233} & & & & & & & & \\ & & & & & & & D_{121121} & D_{121332} & D_{121233} & & & & & & & & \\ & & & & & & & & D_{332332} & D_{332233} & & & & & & & & \\ & & & & & & & & & D_{233233} & & & & & & & & \\ \hline & & & & & & & & & & D_{333333} & D_{333113} & D_{333131} & D_{333223} & D_{333232} & & & \\ & & & & & & & & & & & D_{113113} & D_{113131} & D_{113223} & D_{113232} & & & \\ & & & & & & & & & & & & D_{131131} & D_{131223} & D_{131232} & & & \\ & & & & & & & & & & & & & D_{223223} & D_{223232} & & & \\ & & & & & & & & & & & & & & D_{232232} & & & \\ \hline & & & & & & & & & & & & & & & D_{231231} & D_{231123} & D_{231132} \\ & & & & & & & & & & & & & & & & D_{123123} & D_{123132} \\ & & & & & & & & & & & & & & & & & D_{132132} \end{array} \right)$$

Fig. B.3. The structure of tensor D.

Data availability

Data will be made available on request.

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